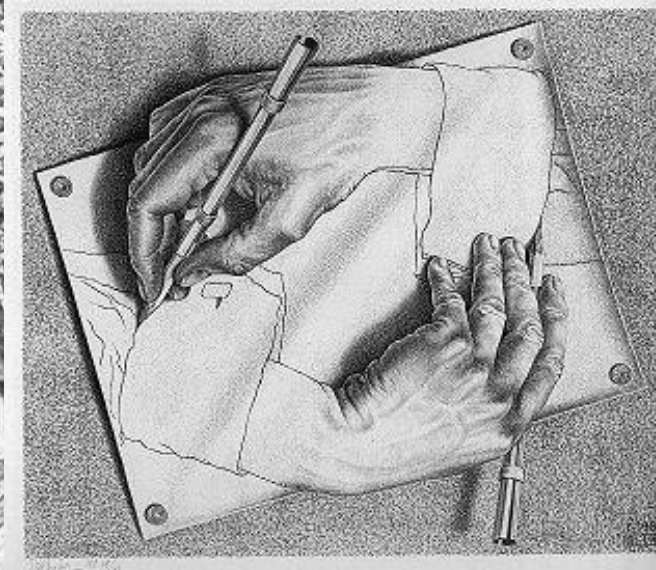
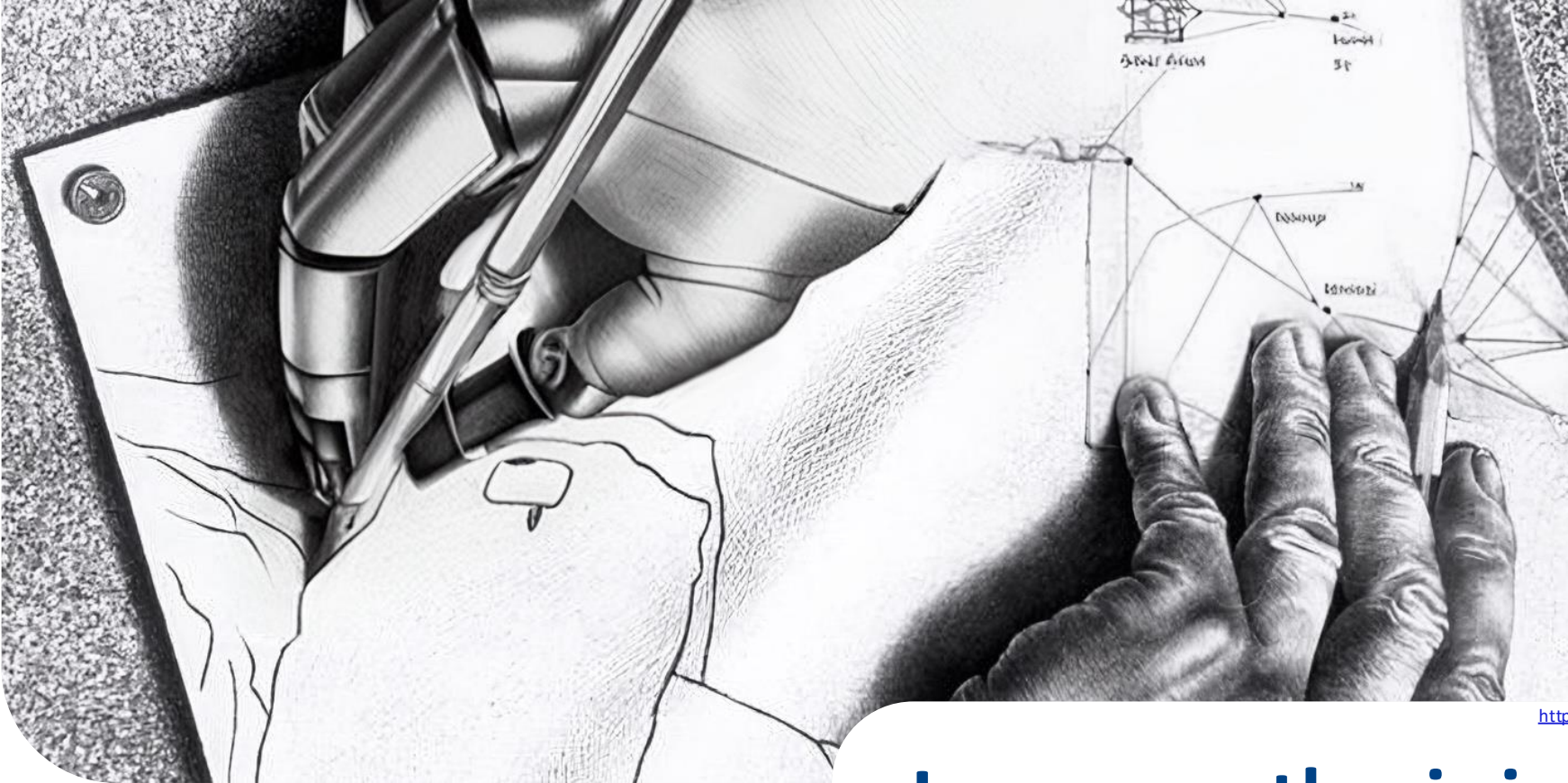




Drawing Hands
M. C. Escher
1948

<http://www.andrewkrill.com/index.php/2022/12/14/diffusing-hands/>

Image synthesis in radiotherapy

State-of-the-art, clinical integration and future directions

MIART @ MICCAI 2026 · Strasbourg · 1 October 2026



Matteo Maspero
Assistant professor, Med phys
UMC Utrecht, Center for Image Sciences
Radiotherapy Department
m.maspero@umcutrecht.nl



AlinRT 2027

Artificial Intelligence in Radiotherapy



PEER-REVIEWED INTERNATIONAL SYMPOSIUM

1–2 April 2027



UMC Utrecht & Princess Máxima Center · Utrecht, the Netherlands

Abstract submission closes **1 December 2026**

Six 90-min sessions · Invited state-of-the-art talks (30 min) · Proffered papers (9+3 min)
A cross-disciplinary keynote each day · Double-blind review

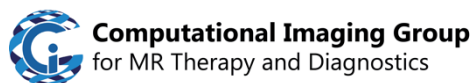
Sessions

- S1** Segmentation & Registration
- S2** Reconstruction & Synthesis
- S3** Foundation Models, Text, Explainability & Uncertainty
- S4** Dose & Adaptive Workflows
- S5** Clinical Predictions & Outcomes
- S6** Implementation, QA & Ethics



<https://AlinRT.org> — Call for Abstracts opens on 1 September

Organized by



In collaboration with



A disclaimer

on what I think a keynote is for

I did not come here to give you answers, but to raise questions

Generalist models can now produce a fluent, plausible answer to almost any question you put to them.

So the scarce skill is no longer producing the answer, but knowing which question is worth asking

→ and noticing when a confident answer is the wrong one

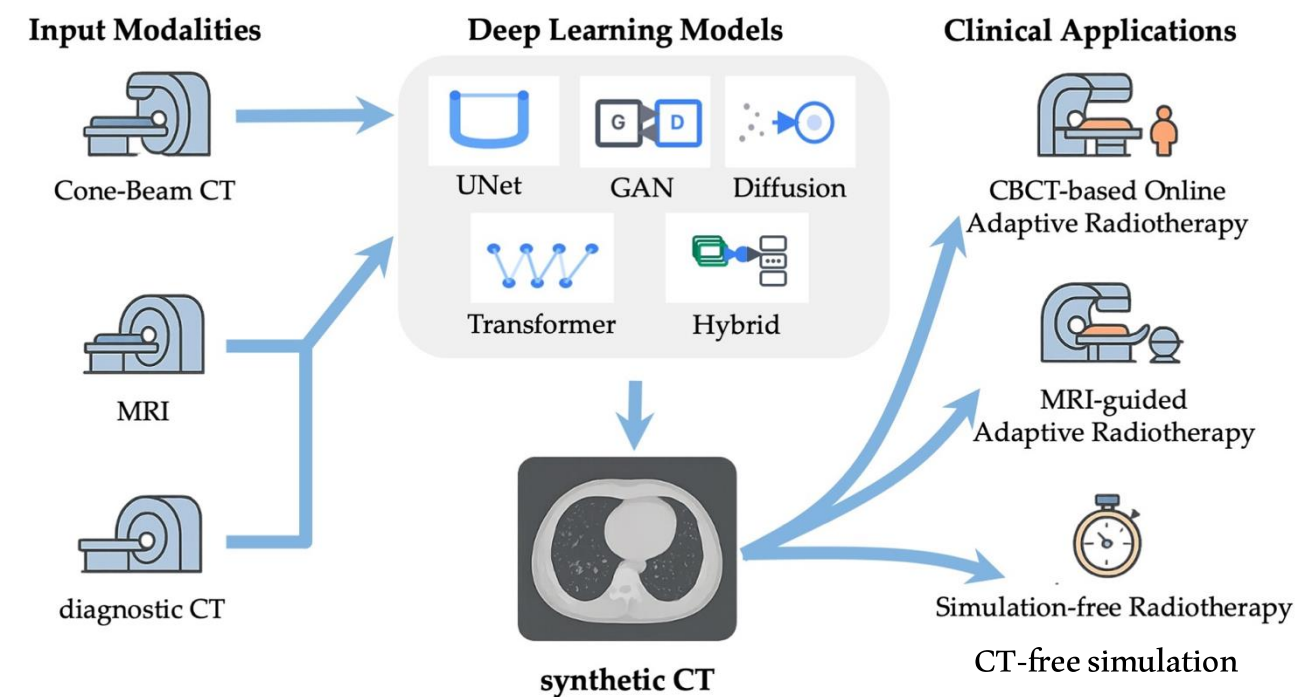
I will phrase four questions today

Whether they are the sensible ones is genuinely for you to judge

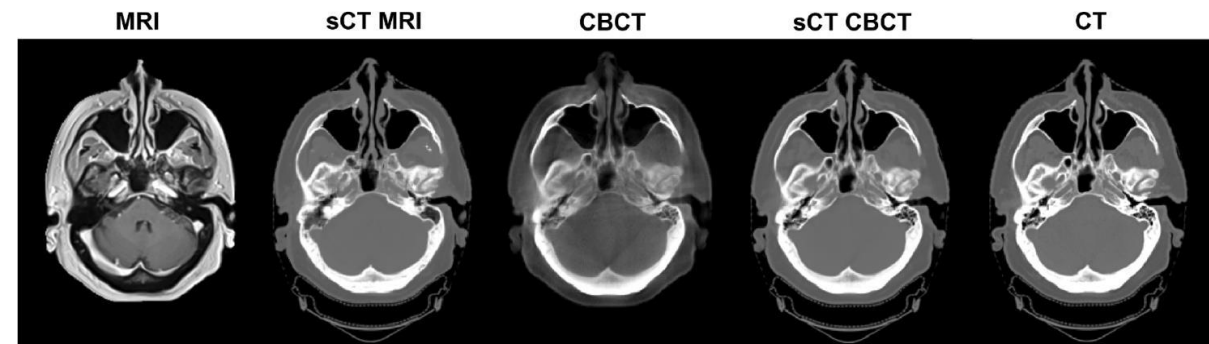
(Tell me in the discussion if I picked wrong)

Main application of synth

→ RT



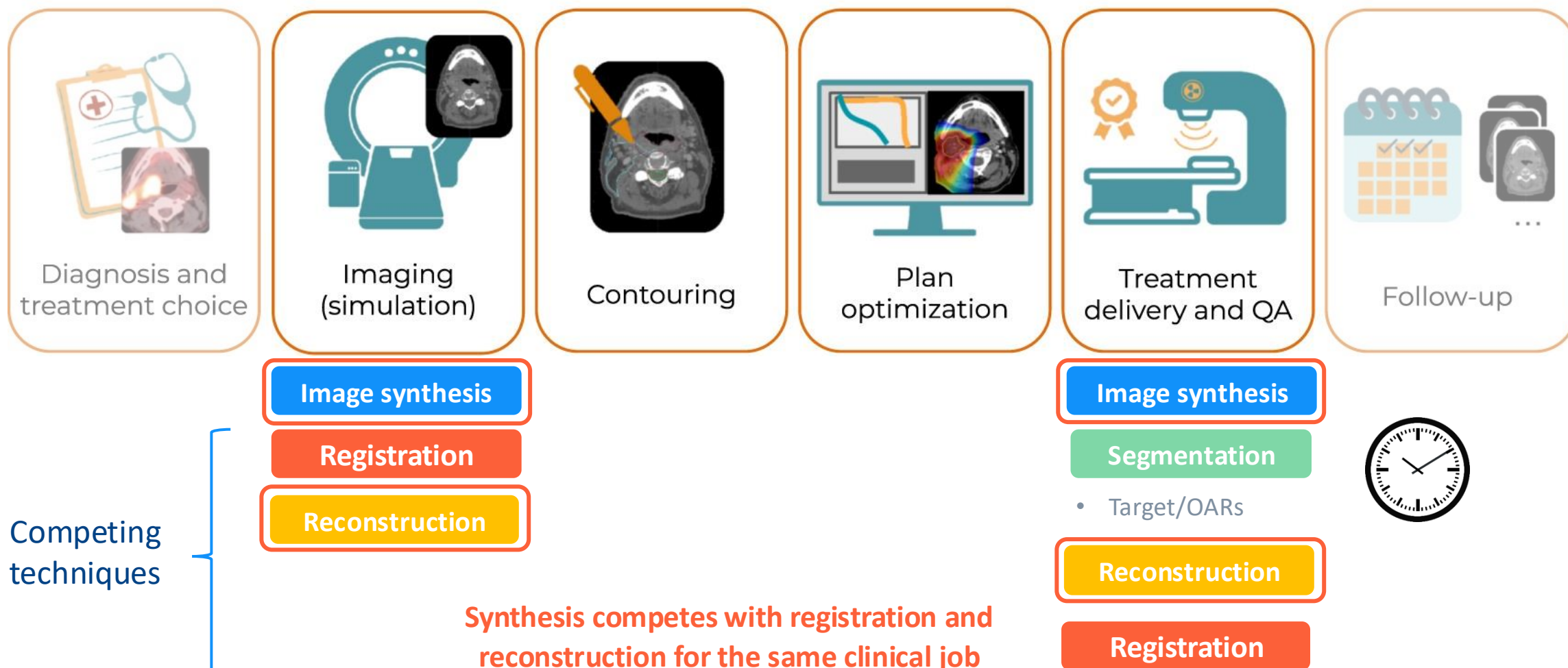
- Target modality is mostly CT
→ importance of CT for treatment planning and dose calculation
- Common source modalities:
→ From MRI and CBCT to synthetic CT



Landry et al. 2024 <https://doi.org/10.1140/epjp/s13360-024-05660-8>

- Other “exotic” applications, currently less adopted

The RT workflow & opportunities for synth/recon



One object, five names

nomenclature

synthetic CT

pseudo-CT

virtual CT

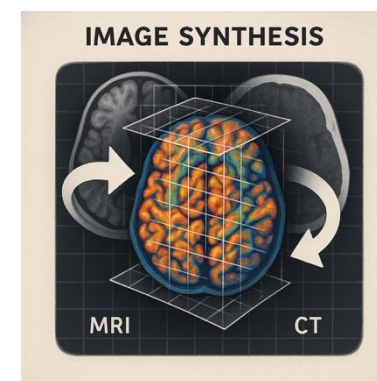
substitute CT

corrected CBCT

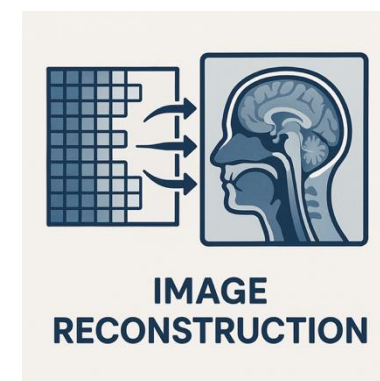
No agreed convention — and the names hide different mechanisms

- image synthesis: a learned intensity mapping
- deformable registration of the planning CT: no synthesis at all
- scatter/artefact correction in the reconstruction: also not synthesis

The same product name can mean any of the three: ask which one



synthesis
image → image



reconstruction
raw data → image

In this talk:

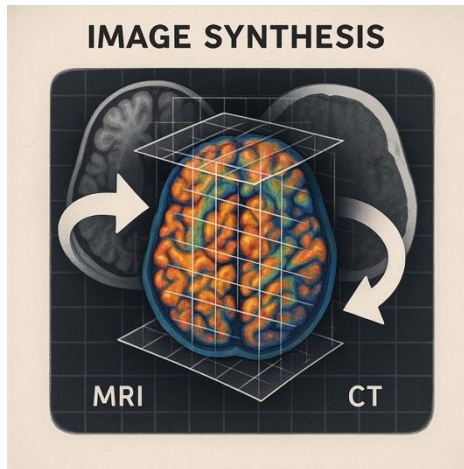
sCT = image synthesis and post-processing only

Definitions

Image synthesis

Generation of medical images from text, same or other modalities, anatomical priors, or learned representations

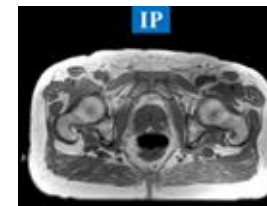
→ ...-to-image translation



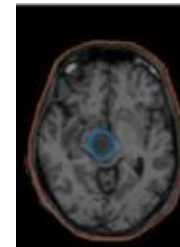
Using image synthesis requires a leap of faith

→ Before going clinical
→ Do I trust a method? (why?)

1. MRI-to-CT

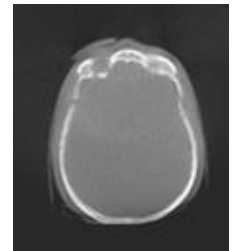
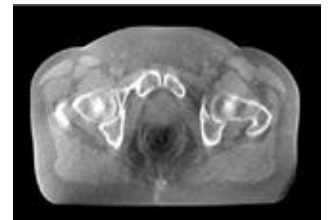


Pelvis



Brain

2. CBCT-to-CT



What is shocking about image synthesis...

The most important image for a patient's treatment becomes one no scanner ever acquired

MRI-only RT

plan on MRI
no planning CT

CBCT-based ART

adapt on the
image of the day

Simulation-free

no simulation
session at all

routine in a growing number of departments — not a proposal

Four questions

35 minutes

1

Is it good enough?

The technical arc, and the four things medical data adds to it

2

Is it in the clinic?

Benchmarks, regulation, and what is actually running at 8 a.m.



3

What are we trying?

Prior work on every problem I will have just complained about

Slides available until 30
October at
<https://surfdive.surf.nl/s/Aago8teqPRp7jep>

4

What comes next?

Four bets, in decreasing order of confidence

A good chunk of this room already builds these models, so I will spend the time on what the papers leave out

1

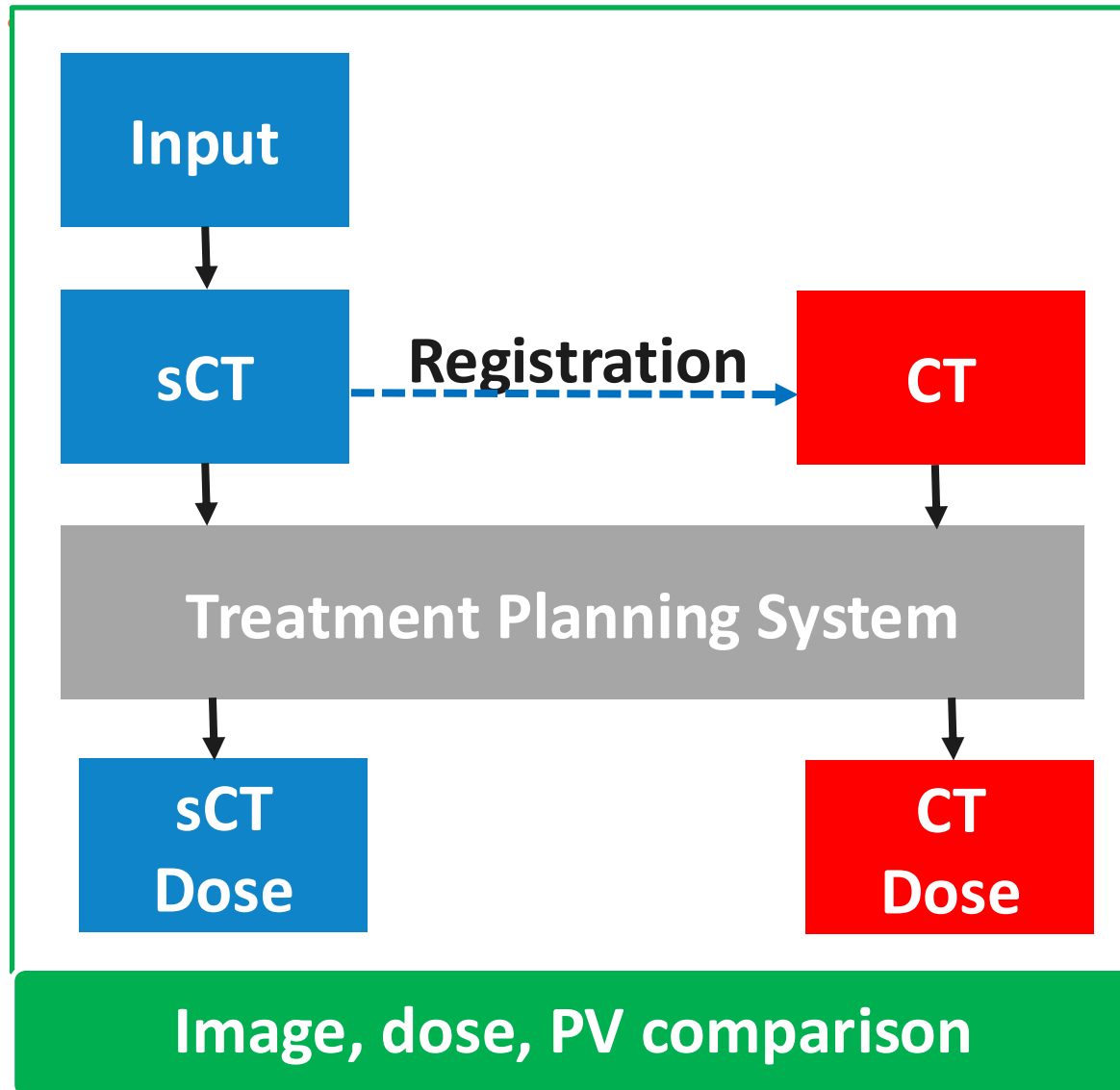
QUESTION 1 — TECHNICAL STATE OF THE ART

Is it good enough?

The architecture story you already know, and the four constraints that medical data adds to it

≈ 10 minutes

How to check the sCT?



→ Which registration?

- **Image-similarity**

Mean Absolute Error

$$MAE = \sum_{i=1}^N \frac{|sCT_i - CT_i|}{N};$$

Mean Error

$$ME = \sum_{i=1}^N \frac{sCT_i - CT_i}{N}$$

with i = voxels in body

(R)MSE

PSNR

SSIM

Recommendations^{1,2}

no guidelines

- **Geometry**

- Body/other contours differences
- Anatomical landmarks
- Structure segmentation, e.g. bones

- **Dose accuracy**

- Dose differences in VOI
- Dose Volume Histograms
- Gamma Analysis

- **Position verification accuracy**

- Matching

Spadea MF & Maspero M et al. [Med Phys](#) 2021; 48(11).

Paganelli C, et al. [Med Phys](#). 2018 Oct;45(10):e908-22.

¹Vandewinckele L, et al. [Radiother Oncol](#). 2020 153, 55-66. ²NCS report 36 MRI QA in RT

Twenty years in one line

you know this curve



Each step bought fidelity, none of them bought generalisation

The 2017 model and the 2026 model may fail on the same thing: a scanner they have not seen

What each family buys you

and what it costs

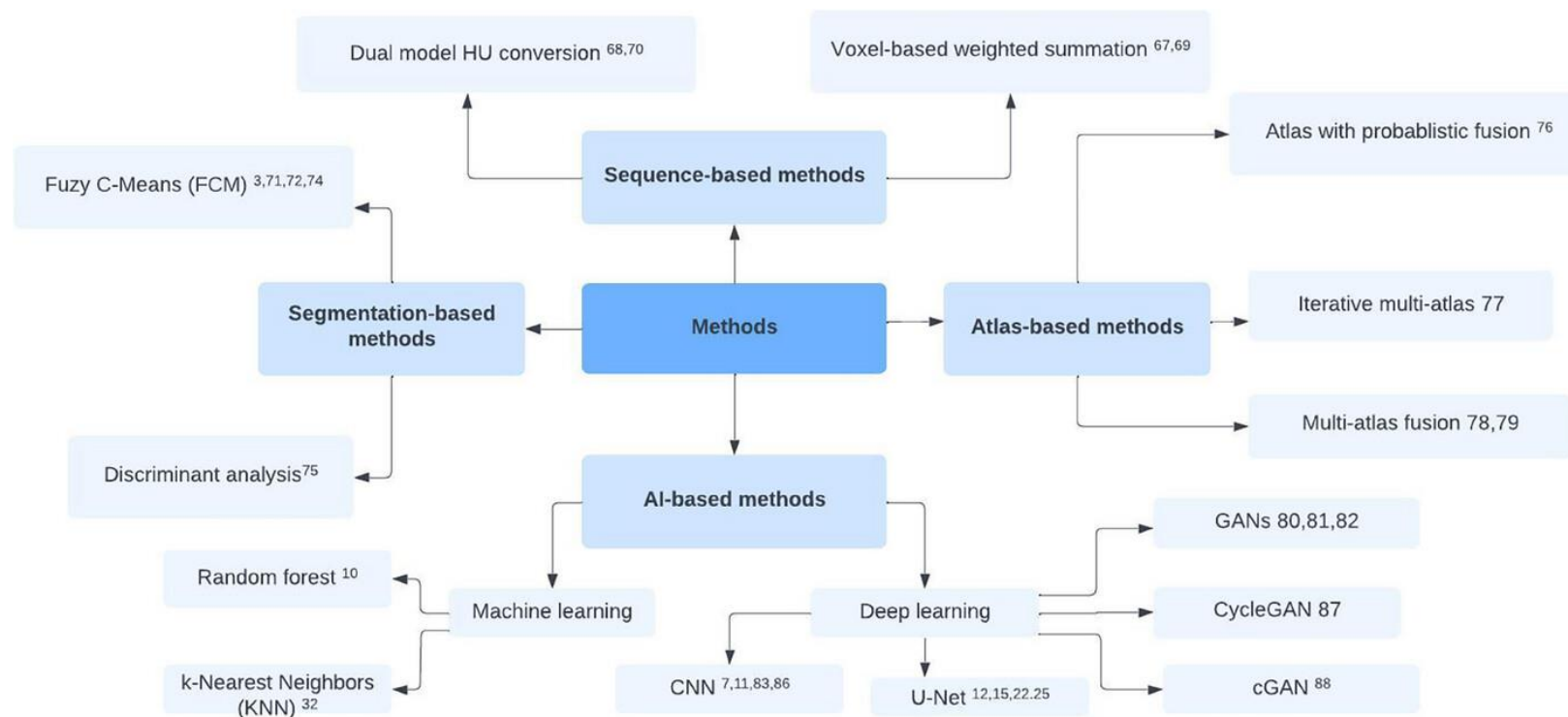
From: sCT and Dose Calculation

	Bulk density assignment	Atlas-based approaches	Voxel-based approaches	
			Statistics	Machine learning
Definition	Assignment of homogeneous densities to MRI segmented classes	Performing deformable registration of the given MRI to one or several co-registered MRI-CT atlas(es)	Use of voxel intensity information for sCT prediction	
Pre-requisite	Segmentations on MRI + density values per structure	Atlas database (MRI-CT pairs)	Statistic model definition	Training step to establish the MRI to CT translation model
Advantages	• Simple • Fast	• Continuous values of HU	• Continuous values of HU if regression	• Continuous values of HU • Fast • Does not require access to matched data for the latest deep-learning methods
Inconvenients	• Need of specific MR sequence(s) for bone delineation • Homogeneous areas in the sCT	• Dependent on deformable registration accuracy • Calculation time (depending on the number of patients considered in the database) • Not robust to atypical patient anatomies	• Potential need of specific MR sequence(s) enhancing bone visibility • Model performance dependency	• Potential need of specific MR sequence(s) enhancing bone visibility • Not robust to atypical patient anatomies (if not present in training data) • Learning step computationally costly

Every row in the last column is a generalisation failure mode

Before deep learning

and it already worked



Bulk density and atlas methods reached < 2 % dose difference in brain and pelvis, before any network

The bar deep learning had to clear was never 'produce a CT'. It was 'produce one that generalizes and is fast'

Where the field landed

MRI-to-CT 51 studies

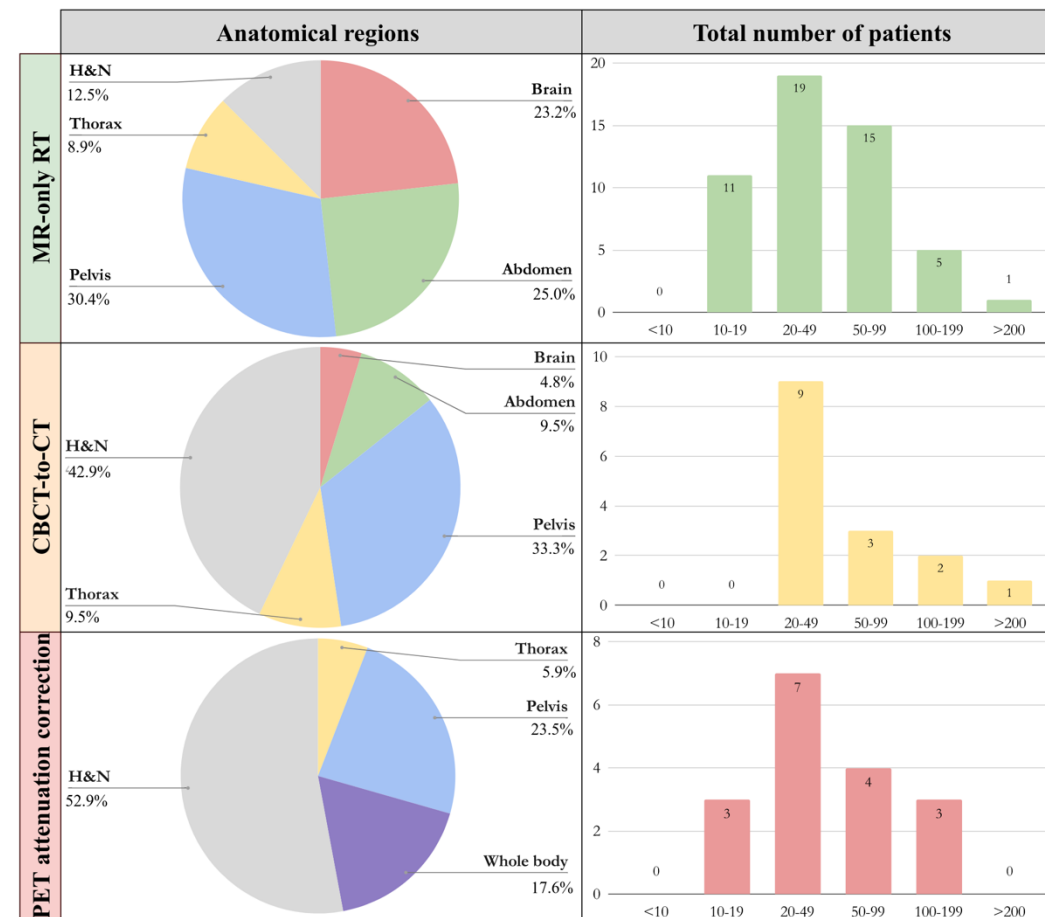
- GANs 55 % · U-Net 35 % · other CNNs 10 %
- dose difference < 1 %, photons and particles
- only 4 multi-centre

CBCT-to-CT 31 studies

- cycle-GANs ~60 % · U-Net ~25 %
- 2D majority; only 6 studies fully 3D

Accuracy was solved on single-centre data. That is the whole caveat.

systematic review, 2021



You know the architectures...

That was not the hard part

What follows is what medical data does to all of them:
four constraints that do not appear in natural-image synthesis

Constraint 1 = clinical objectives

a photorealistic sCT can be dosimetrically wrong

What the loss optimizes = image similarity

per-voxel intensity agreement

perceptual / adversarial realism

MAE, PSNR, SSIM



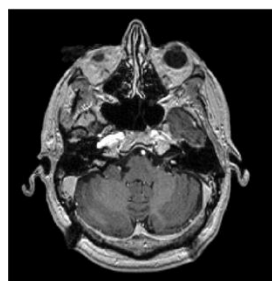
What the patient needs: dose at the right spot

Accurate HU → correct stopping power

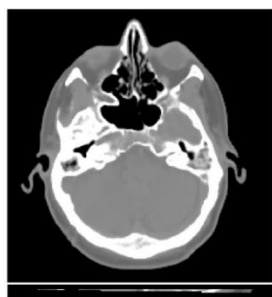
Correct representation of tissue interfaces, where dose gradients live

Geometric fidelity

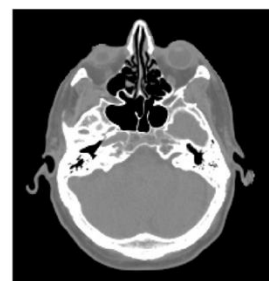
These two are correlated but they are not the same objective, and nothing in the training loop enforces the right-hand one



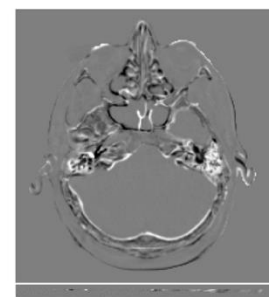
MR



sCT



Real CT



Difference Map



MIART TODAY

Fernandez Salamanca et al. — Recognizing Dosimetrically Irrelevant Edits of Auto-Segmentations in Head and Neck Organs of Interest — Oral 1, and the right question: which errors are dosimetrically irrelevant?

Constraint 2 = tricky plausibility

hallucination is a safety problem, not an aesthetic one

A natural-image model that invents plausible texture is doing its job

A sCT that invents plausible anatomy has produced a confident, well-formed, wrong image

Why it is worse here

- the failure is locally plausible — it looks like anatomy, so visual QA does not catch it
- it is not flagged: the model has no way to say 'I have not seen this'
- adversarial and reconstruction losses actively reward plausibility
- 3D consistency is not enforced by 2D or patch-wise training

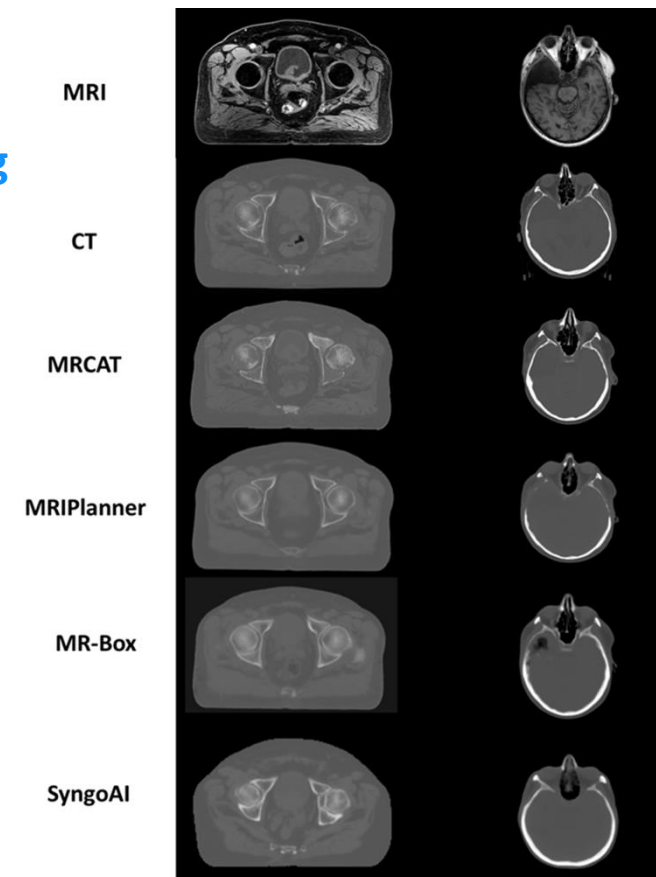
Per-voxel uncertainty is the missing output: nothing deployed produces one

MICCAI 2026

T-AM-093 · Zala et al. — Attention is Matter for Inclusiveness: Generating Synthetic CT for Patients with Hip Implants — the failure mode made explicit: metal implants, where plausible is definitely not correct.



Antun V, et al. Proc Natl Acad Sci USA 2020;117(48):30088-95. <https://doi.org/10.1073/pnas.1907377117> — instabilities in learned reconstruction
Cusumano D, et al. Radiother Oncol 2026;111530 (MESCAL). <https://doi.org/10.1016/j.radonc.2026.111530> · Vandewinckele L, et al. Radiother Oncol 2020;153:55-66.
<https://doi.org/10.1016/j.radonc.2020.09.008>



real MRI, real CT, and four cleared sCT products.
Every one of them reads as a CT

Constraint 3 = underlying error in GT

the ground truth is not ground truth

The target CT and the source MRI/CBCT are acquired at different times

- bladder and rectal filling differ
- bowel gas moves
- patient position and immobilisation differ
- weight and anatomy change between scans

So the 'paired' training set is registered, not identical

- residual deformation is baked into the supervision
- the loss penalises the model for the registration's error

A model can only be as consistent as the registration that defined its target

This is also why unpaired training became popular, but it sidesteps the problem rather than solving it

MIART TODAY

Guija-Valiente et al. — SynthRCT: Scalable Conditional Deformation Synthesis for Synthetic Repeat CT Generation — Oral 2: synthesise the repeat CT instead of hunting for one.

Constraint 4 = generality

the actual open problem

scanner

vendor, field strength,
coil, reconstruction kernel

protocol

sequence, TE/TR, FOV,
resolution, contrast agent

site

anatomy, immobilisation,
patient population

time

protocol drift, software
upgrades, new hardware

MRI is the hard case — it is not quantitative

- the same sequence name gives different contrast on different vendors
- CBCT-to-CT is measurably easier: smaller domain gap; both are X-ray

Single-center accuracy is no longer interesting → resiliency to multi-center data is

MIART TODAY

Rangnekar et al. — Parameter-Efficient Pretrained-CT-to-MRI Transfer for Rectal Cancer Segmentation: Performance-Calibration Trade-Offs — Oral 1: transfer, and what it costs you in calibration.

T-PM-123 · Liu et al. — IntraStyler: Intra-Domain Style Synthesis for Cross-Modality MRI Domain Adaptation · T-AM-109 · Iele et al. — Multi-Agent Test-Time Adaptation for Robust Medical Image-to-Image Translation

So we built a benchmark

SynthRAD



<https://synthrad2023.grand-challenge.org>

<https://github.com/SynthRAD2023>



<https://synthrad2025.grand-challenge.org>

<https://github.com/SynthRAD2025>

SynthRAD2023

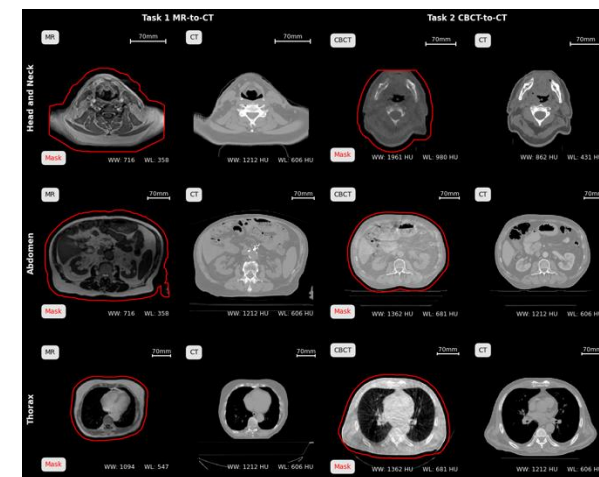
- 1080 datasets · 3 centres · brain and pelvis
- MR→CT and CBCT→CT

SynthRAD2025

- 2362 datasets · 5 centres · head & neck, thorax, abdomen
- the anatomies everyone had been avoiding

Data, pre-processing, metrics, and evaluation are all open

SynthRAD2023 <https://synthrad2023.grand-challenge.org>
SynthRAD2025 <https://synthrad2025.grand-challenge.org>



Task 1 MR→CT
Task 2 CBCT→CT



UMC Utrecht



umcg



UNIKLINIK
KÖLN

Radboudumc



dkfz.

GERMAN
CANCER RESEARCH CENTER
IN THE HELMHOLTZ ASSOCIATION

What the benchmark actually taught us

CBCT→CT beats MRI→CT on similarity and dose, as the domain gap predicts

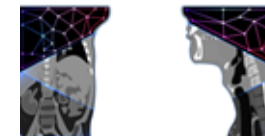
Brain, pelvis, and head & neck are ahead of thorax and abdomen

Clinically acceptable for photons; protons still need work

CNN and GAN backbones still lead: diffusion or other architectures have not swept the board

SynthRAD

**SynthRAD
2025**



**SynthRAD
2023**



Architecture choice explained less of the variance than data and pre-processing did

For a room full of method builders, that is the uncomfortable result, but a useful one

Take away 1

The architecture race is largely over, and it is not what limits us

Four medical-specific constraints do:

- dosimetric $\neq$ perceptual accuracy
- hallucination is a safety failure with no uncertainty output
- the supervision target is itself misregistered
- domain shift across scanner, protocol, site and time

Generalisation, not fidelity, is the open technical problem

2

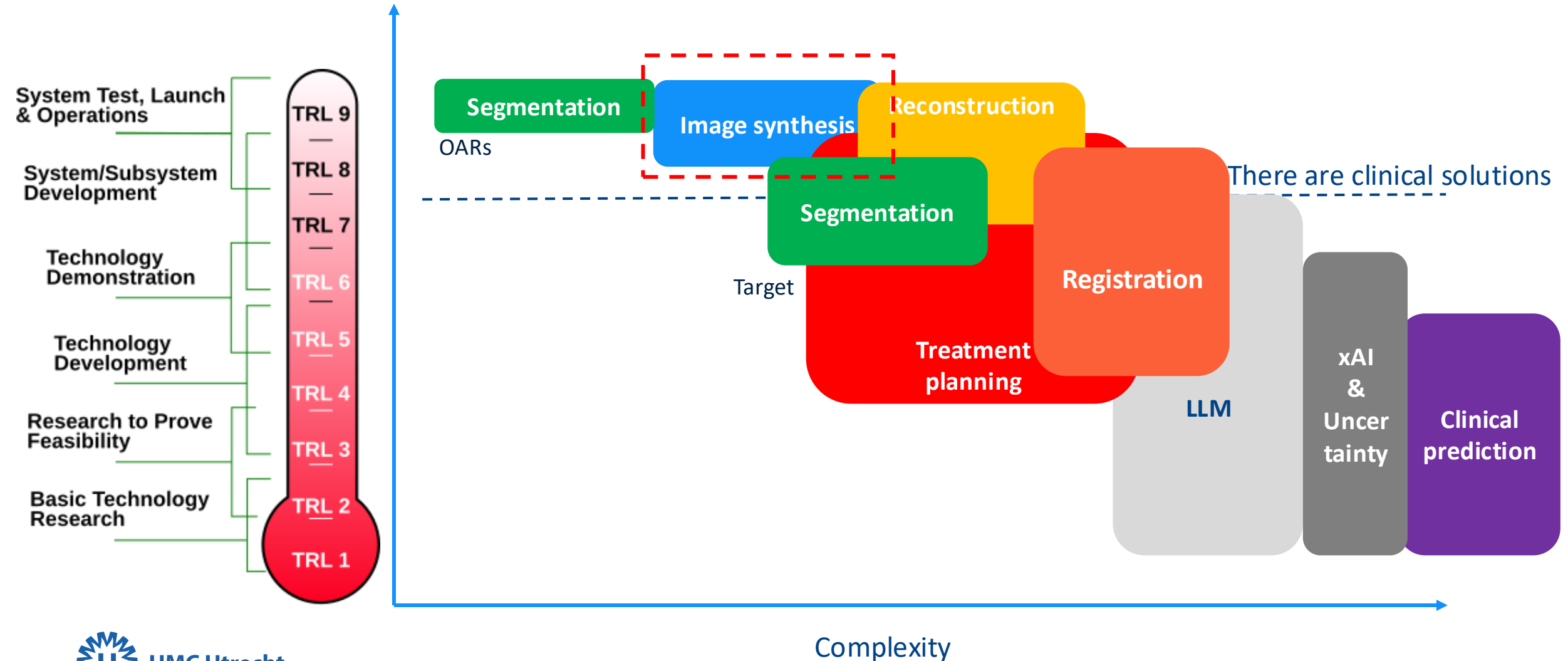
QUESTION 2 — CLINICAL INTEGRATION

Is it in the clinic?

The part you do not get from a methods paper: what counts as evidence, what is cleared, and what is actually running

≈ 9 minutes

Technology readiness level



Commercial solutions so far...

[2026-09]



Product	Company	Modality	CE	FDA	Anatomy	Link
MRCAT Brain	Philips	MRI	cleared	510(k) Cleared	Brain	Link
MRCAT Pelvis	Philips	MRI	cleared	510(k) Cleared	Pelvis	Link
MRCAT Prostate + Auto-Contouring	Philips	MRI	cleared	510(k) Cleared	Pelvis	Link
MRCAT Head and Neck	Philips	MRI	cleared	510(k) Cleared	Head, Neck	Link
syngo.via RT Image Suite	Siemens Healthineers	MRI	cleared	510(k) Cleared	Brain, Pelvis	Link
MRiplanner	Spectronic Medical (GE)	MRI	cleared	510(k) Cleared	Pelvis, Brain, Head & Neck	Link
MR-Box (Art-Plan+)	Therapanacea	MRI	cleared	510(k) Cleared	Pelvis, Brain, Head & Neck	Link
AdaptBox (Art-Plan+)	Therapanacea	CBCT	cleared	510(k) Cleared	Head & Neck, Thorax, Breast, Pelvis	Link
Workspace+ (Image+)	MVision AI	MRI, CBCT, CT	cleared	510(k) Cleared	Brain, Pelvis, Pelvis, Head & Neck, Thorax, Abdomen	Link
MOZI TPS	Manteia	CBCT	cleared	510(k) Cleared	All sites	Link



Bahloul MA, et al. Advancements in synthetic CT generation from MRI: A review of techniques, and trends in radiation therapy planning. J App Clin Med Phys. 2024;25(11):e14499 <https://doi.org/10.1002/acm2.14499>

What is actually cleared

not what is published

6 companies
offering sCT

10 products
all AI-based now

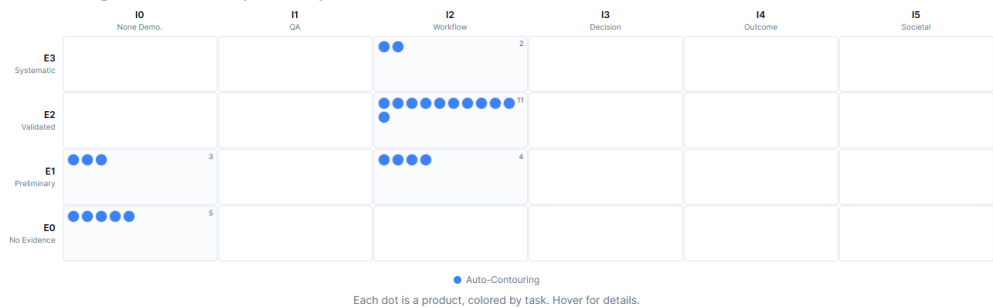
11 products
for reconstruction



an open, maintained inventory of CE / FDA-cleared deep-learning products in radiotherapy

<https://dlinrt.eu>

Evidence Rigor vs Clinical Impact (25 products)



evidence rigour (E0–E3) vs clinical impact (I0–I5), all products

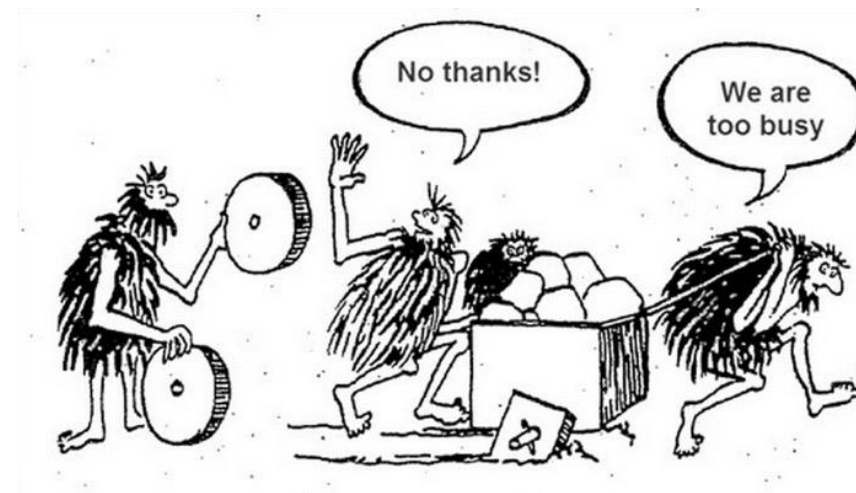
The cleared sCT products

Six companies, ten product: all AI-based

- two integrated with an MRI vendor; the rest standalone or cloud
- brain, pelvis, head & neck
- nothing cleared for thorax or abdomen

What none of them ship

- a per-voxel uncertainty or confidence map
- a documented out-of-distribution behavior
- cross-vendor validation on scanners they were not trained on



No tools to facilitate commissioning

- Clinics are left to re-invent the wheel every time
- Real bottleneck for diffusion of technology

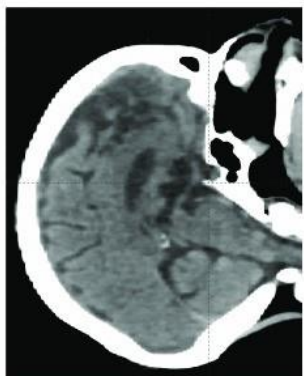
The regulatory file certifies the intended use. It does not certify that your scanner is inside it.

Comparison 4 sCT solutions

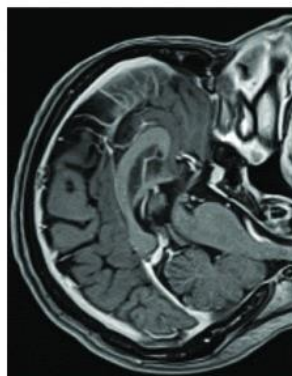
1/2

Brain

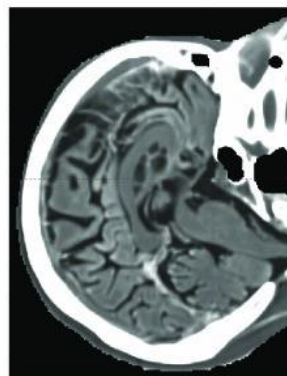
CT



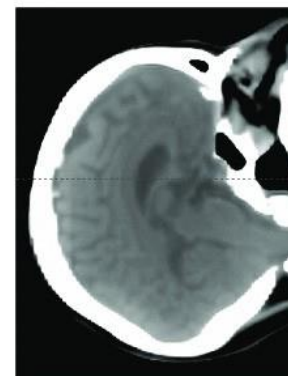
Delineation MR



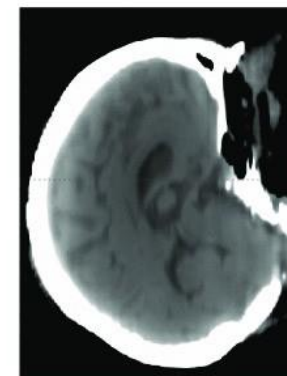
sCT Syngo_BD



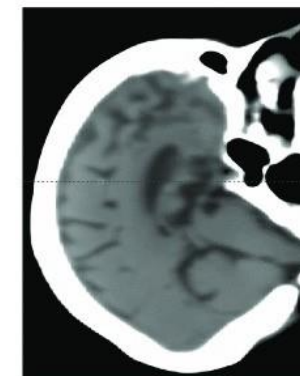
sCT Spectronic



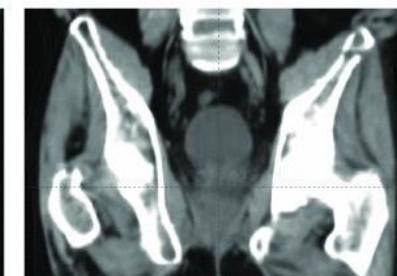
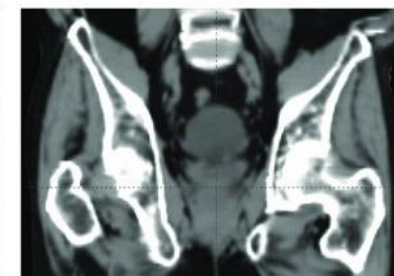
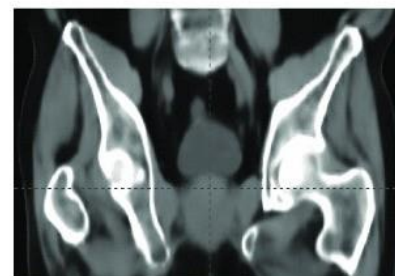
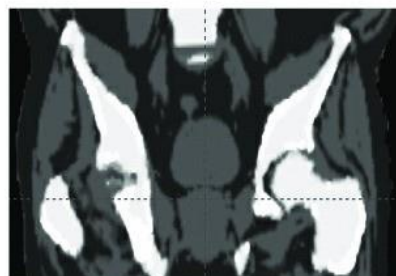
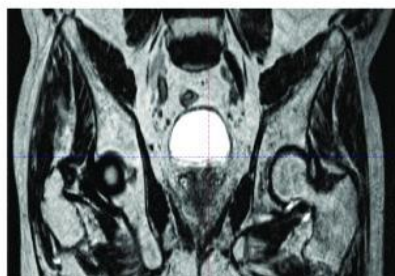
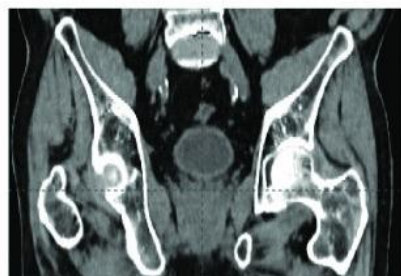
sCT Syngo_AI



sCT Therapanacea



Pelvis



SIEMENS
Healthineers

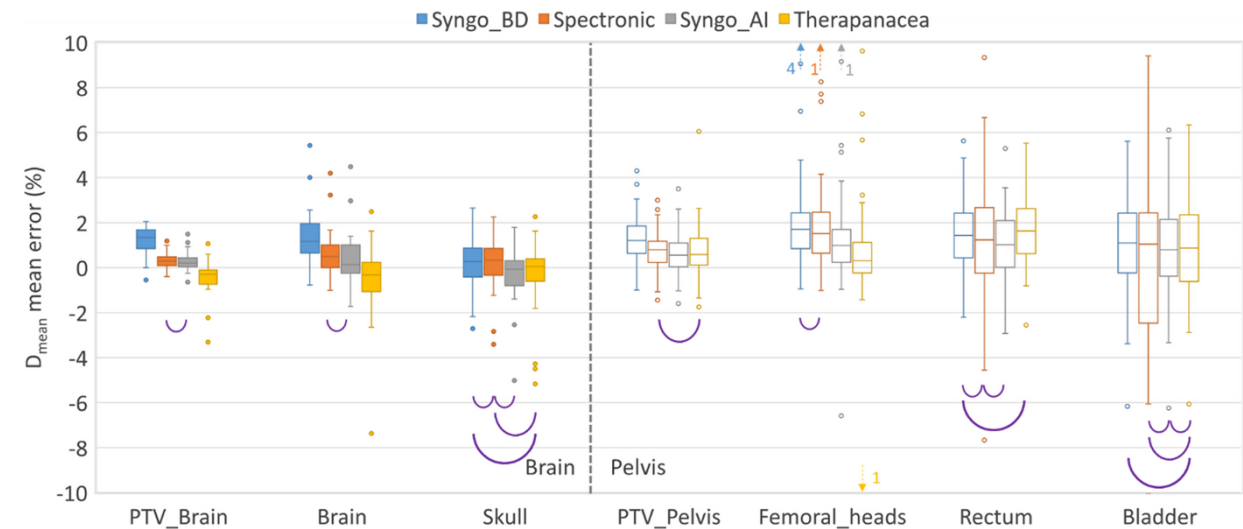
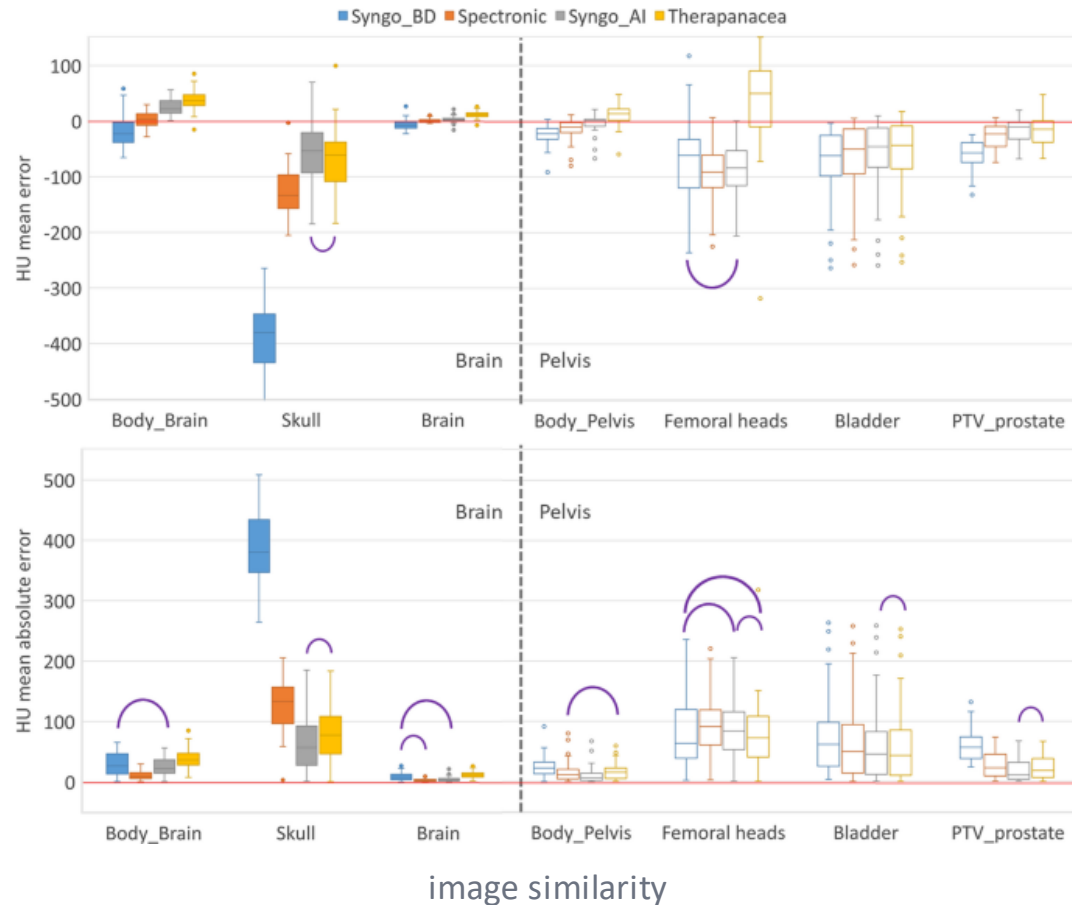
SPECTRONIC
MEDICAL

SIEMENS
Healthineers

THERAPANACEA

The metrics

four cleared products, same patients



dose difference

No clear winner: Mean dose differences stay under 2 % across all four; the similarity ranking does not predict it

Visual analysis

No detection of air bubbles
Difference in HU - spongy bone

Blurred image

No detection of air bubbles
Blurred image

Blurred Image
Difference in HU - caudal bone

*On a different casus

MRI

CT

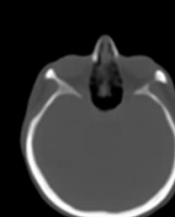
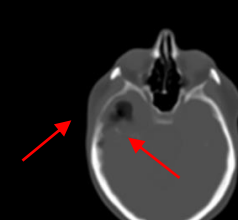
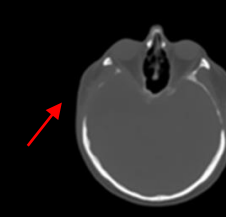
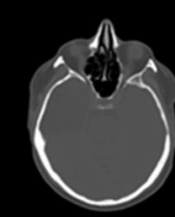
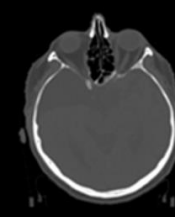
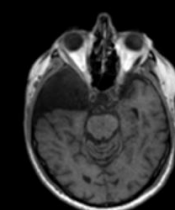
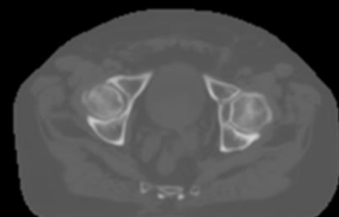
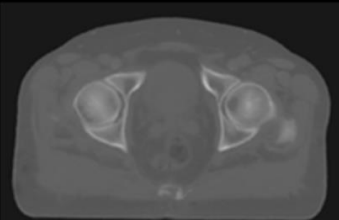
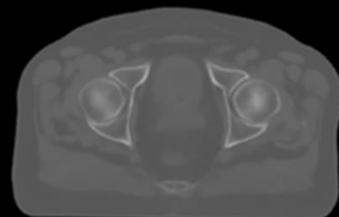
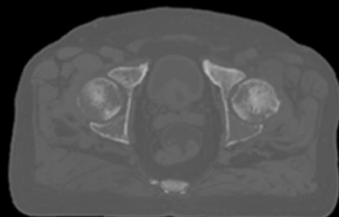
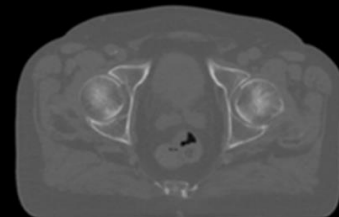
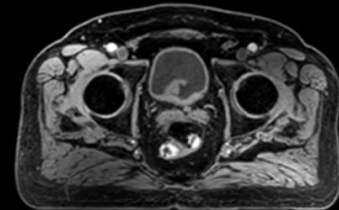
MRCAT

MRIPanner

MR-Box

SyngoAI

*



Overall/ first impression
~ likely to be CT

Skull HU accuracy

Skull HU accuracy
Blurred image

Skull HU accuracy

Skull HU accuracy
Blurred image

Acceptance ranges

Population-based ranges over all the commercial solutions, outlier cases excluded

Index	Acceptance ranges			
	Optimal range		Mandatory range	
	Brain	Pelvis	Brain	Pelvis
MAE (body)	<109 HU	<51 HU	<118 HU	<54 HU
MAE (bone)	<289 HU	<238 HU	<307 HU	<283 HU
ME	-28 HU<ME<12 HU	-9 HU <ME<17 HU	-31 HU <ME<13 HU	-19 HU <ME<26 HU
SSIM	>0.67	>0.72	>0.65	>0.70
PSNR	>26 dB	>33 dB	>25 dB	>32 dB
D (PTV) (%)	-0.4% <D<1.7%	-0.6% <D<1.1%	-0.6% <D<1.8%	-1% <D<1.3%
D (OAR) (%)	-1.1% <D<0.9%	-0.2% <D<0.9%	-1.7% <D<1.7%	-0.3% <D<1.3%
Gamma Pass Rate	>99.7%	>99.2%	>99.6%	>98.7%
Left-Right (x)	-0.8 mm<x<0.8 mm	-1.1 mm<x<0.8 mm	-1.2 mm<x<1.1 mm	-1.7 mm<x<1.2 mm
Antero-Posterior (y)	-1.2 mm<y<0.5 mm	-0.7 mm<y<1 mm	-1.6 mm<y<0.7 mm	-1.2 mm<y<1.5 mm
Superior-Inferior (z)	-0.4 mm<z<1.0 mm	-0.4 mm<z<2.7 mm	-0.6 mm<z<1.1 mm	-0.9 mm<z<3.4 mm
Tilt (T, °)	-0.6°<T<1.2°	-0.6°<T<0.7°	-1°<T<1.5°	-0.9°<T<1.1°
Pitch (P, °)	-0.5°<P<0.2°	-0.3°<P<0.4°	-0.7°<P<0.6°	-0.5°<P<0.5°
Roll (R, °)	-0.4°<R<0.3°	-0.7°<R<0.5°	-0.6°<R<0.6°	-0.8°<R<0.8°

Pragmatic rule of thumbs
<55 HU
-25 HU <ME<25 HU
>0.65/0.70
>25/32 dB
D _{PTV} <1%
D _{OAR} <1%
>98.5%
x <1 mm
y <1 mm
z <2 mm
T <1°
P <1°
R <1°

Tolerance ranges for each evaluated metric, reported separately for brain and pelvic cases. Values are categorized into optimal and mandatory acceptance levels, including the 80% and 90% of the cases meeting the two acceptance criteria identified: dose percentage difference inferior to 2% and gamma passing rate (2%/2mm) higher than 95%.

Acceptance criteria now exist

MESCAL

A benchmark dataset and population-based tolerance ranges across commercial sCT solutions

- separate levels for brain and pelvis
- two tiers: optimal and mandatory
- anchored to dose difference $< 2\%$ and gamma ($2\%/2\text{ mm}$) $> 95\%$
- stated as the fraction of cases that must pass: 80 % and 90 %

This is what the field was missing: a number a physicist can fail a product against

Still recommendations, not a guideline, and nothing equivalent exists yet for the online adaptive setting

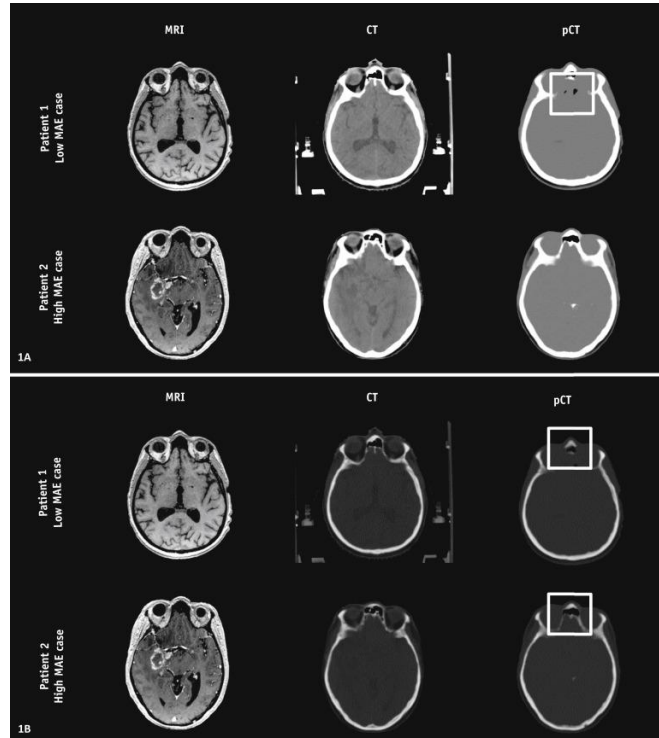
Solving the method reproducibility hurdle?



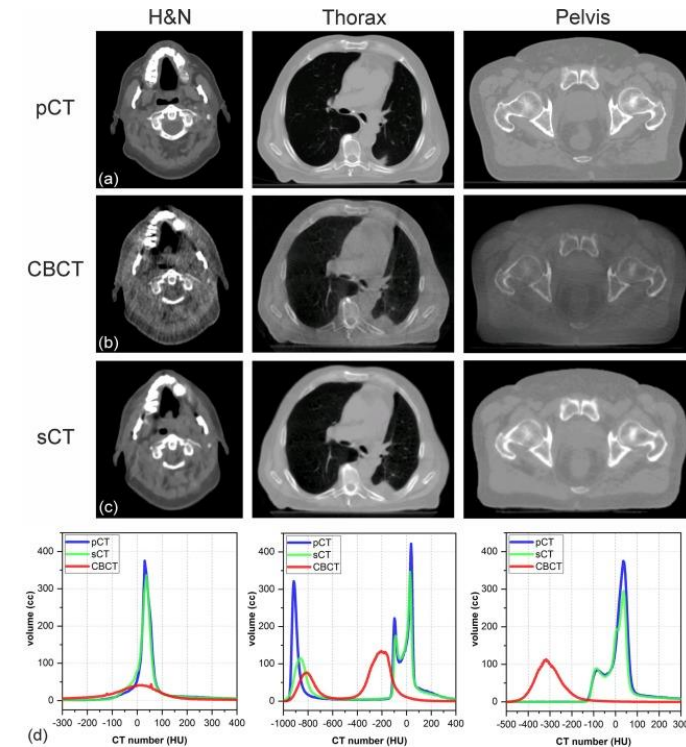
The evidence is real

multi-centre, both modalities

MRI-based · 2 centres · ~200 cases



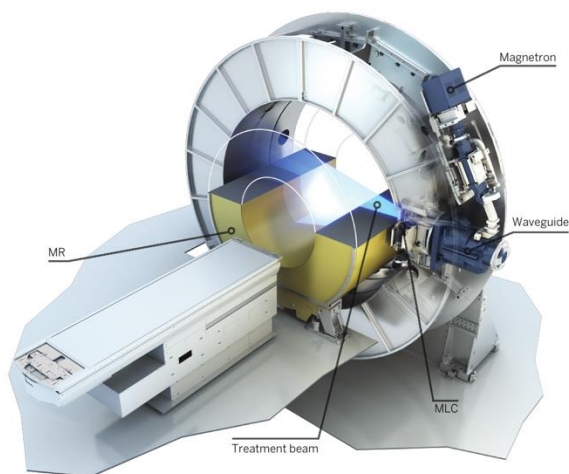
CBCT-based · > 200 cases



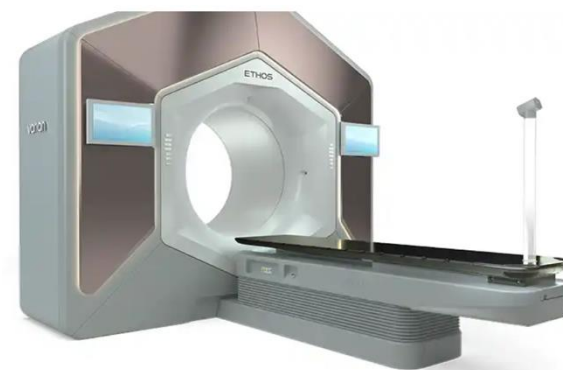
Feasible and accurate to within 1–2 % across centres. Pre-treatment, this question is settled.

From pretreatment to Adaptive (online) scenarios

MRI-based



CBCT-based



Same synthesis techniques
can be applied!



What is actually running

MRI-guided online adaptive

Every commercial MR-linac workflow today

- deformable registration of the planning CT to the daily MRI
- or bulk HU assignment from the planning CT
- or re-delineation on the daily image

None of them operate in a true MR-only workflow

The deep-learning sCT that the literature is full of is not the thing running at 8 a.m. on the treatment floor

What is actually running

CBCT-guided online adaptive

Two routes, and neither is an AI model

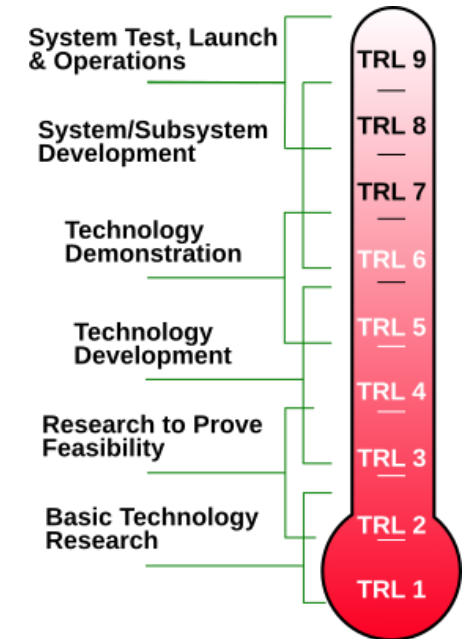
- deform the planning CT onto the daily anatomy — a registration problem
- fix the CBCT itself: larger FOV, scatter correction, HU calibration in hardware and reconstruction

Better hardware and better reconstruction are beating synthesis to the clinic

They are easier to commission, easier to explain, and they fail in ways a physicist already knows how to test

What is deployed is the simplest thing that clears the bar

Not the most accurate model. The one whose failure modes a medical physicist can enumerate, test, and sign for.



**technology readiness
synthesis sits low on it**

The last mile

what it takes to run a model on a patient



Someone in the department has to own this technology. In most places that question is still open.

None of these four steps is about model architecture

All four are about being able to predict how the model fails

Take away 2

Pre-treatment sCT is settled: multi-centre, dosimetrically validated, commercially available

Online adaptive is where the clinic is, and synthesis is not what is running there

Acceptance criteria arrived in 2026; guidelines have not

Image-similarity metrics do not predict dosimetric ranking

The last mile is QA and trust, not model quality

3

QUESTION 3 — ATTEMPTS ON THE TABLE

What are we trying?

I have spent twenty minutes complaining. Here is the prior work on every one of those complaints — and how far it actually gets

≈ 9 minutes

Six things people are trying

one per complaint

● **Uncertainty as per-case QA**

constraint 2 — hallucination

● **Let a second task do the testing**

constraint 2 — no per-case check

● **Stop separating synthesis and reconstruction**

constraint 1 — HU accuracy

● **Put physics in the model**

constraints 1 & 4

● **Domain randomisation**

constraint 4 — domain shift

● **Federated learning**

constraint 4 + validation burden

None of them is finished. All of them are more interesting than another architecture.

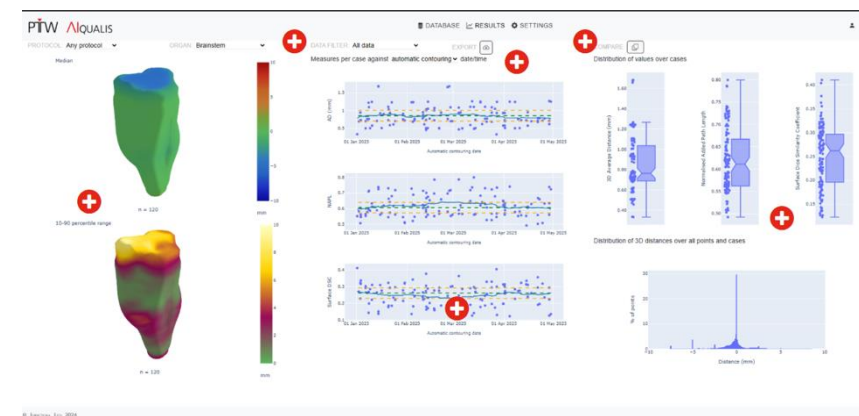
Attempt 1: uncertainty

make the model say when it is unsure

Monte-Carlo dropout and test-time augmentation give a per-voxel uncertainty map essentially for free

And it correlates with the error you care about

- MC dropout uncertainty vs HU error: $r \approx 0.92$
- vs proton range and water-equivalent thickness error: $r \approx 0.6$
- vs dose difference: $r \approx 0.65$
- uncertainty maps are spatially co-located with poor synthesis



the commercial analogue already exists — for contours.
Per-case QA dashboards ship; per-case sCT uncertainty does not.

You can turn that into a DVH confidence interval: a dose band, not a single curve

This is the closest thing the field has to per-case QA, and it is barely used clinically

Ask a vendor for an uncertainty map and watch what happens: is the clinic able to handle this level of complexity / cognitive overload?

MIART TODAY

Rodriguez-Gonzalez et al. — Can We Trust Synthetic CT Algorithms? Uncertainty-Aware Evaluation of CBCT to CT Synthesis for Adaptive Proton Therapy — Oral 2, 10:30. Go to this one.
El Aichi et al. — Uncertainty-Aware Out-of-Field Dose Prediction in External Beam Radiotherapy — Oral 3



Galapon AV, Thummerer A, et al. Feasibility of Monte Carlo dropout-based uncertainty maps to evaluate deep learning-based synthetic CTs for adaptive proton therapy. Med Phys 2024;51(4):2499-509. <https://doi.org/10.1002/mp.16838>
Hémon C, et al. Modeling dose uncertainty in CBCT: predictive approach for deep learning-based synthetic CT generation. Phys Imaging Radiat Oncol 2025;33:100704. <https://doi.org/10.1016/j.phro.2025.100704>
Rusanov B, et al. Phys Med Biol 2024;69(3). <https://doi.org/10.1088/1361-6560/ad1cfc> · PTW AIQUALIS · <https://www.ptwdosimetry.com/en/products/aiqualis>

Attempt 2: second task to check

let a second task do the testing

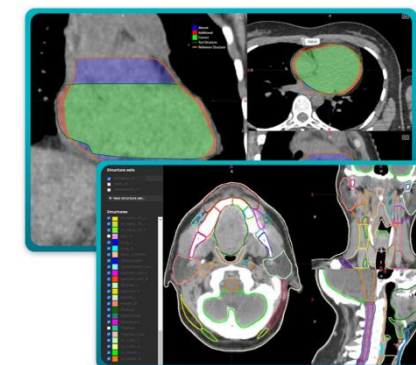
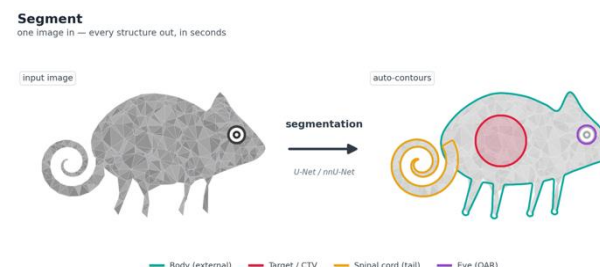
You cannot check a synthetic image against a truth you do not have

But you can check what it lets you do

Segment the synthetic image and test the structures

- are the organs where they should be?
- does the contour agree with the one from the source image?

A downstream task is an observable proxy for a truth you never get



synthesise, then segment

contour QA, in a product

It already works for contours: an automatic contour-QA model with MC-dropout uncertainty reached 93.9 % accuracy at ~13 ms per slice — fast enough for online adaptive.

Same idea, pointed at synthesis. Combine the tasks and the system gets a self-check.

MIART TODAY

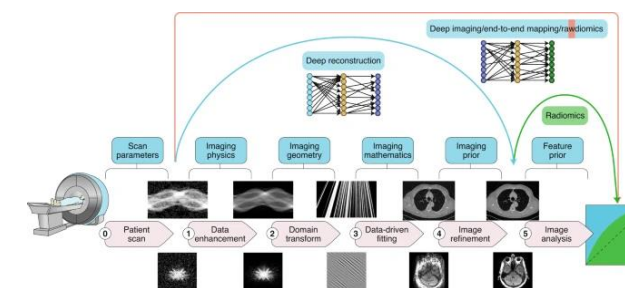
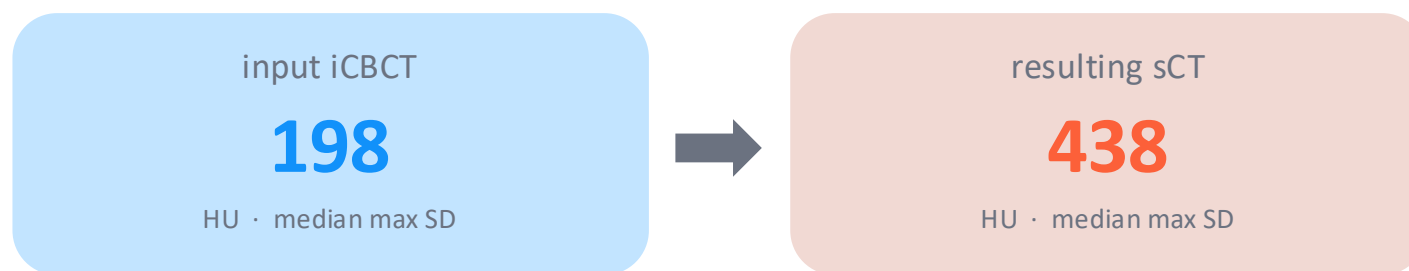
Cubero et al. — CBCT Segmentation in Head and Neck ART: A Longitudinal Deep Learning Approach — Oral 3: the downstream task, run longitudinally on the image of the day.

Attempt 3: combine recon & sCT

stop treating synthesis and reconstruction as separate

A sCT from CBCT inherits everything the reconstruction did first

Vary only the iterative-reconstruction parameters and watch what the synthesis network does with it:



one chain: acquisition → reconstruction → synthesis → analysis.
We commission the middle and the end separately.

The network amplifies a change the eye cannot see in its input. Scatter correction was the worst offender.

So commission the pair, not the model. Re-validate the sCT whenever the reconstruction settings or the scanner software change.

And the honest next step is to stop separating them at all: projection-domain correction is already reviewed as its own class, and volumetric CBCT has been synthesised straight from two projections.

Attempt 4: add physics

put the physics in the model, not just in the loss

The problem with a purely learned mapping is that nothing in it knows what MRI signal is

Physics-informed synthesis

- use the signal equation as part of the network, not as a post-hoc penalty
- predict physical parameters — T1, T2, proton density — then simulate the contrast you want
- generalises to contrasts that were never in the training set

Anatomy-aware losses

- penalise structural implausibility directly, rather than trusting the adversarial term

A parameter map is scanner-independent in a way an intensity map never is

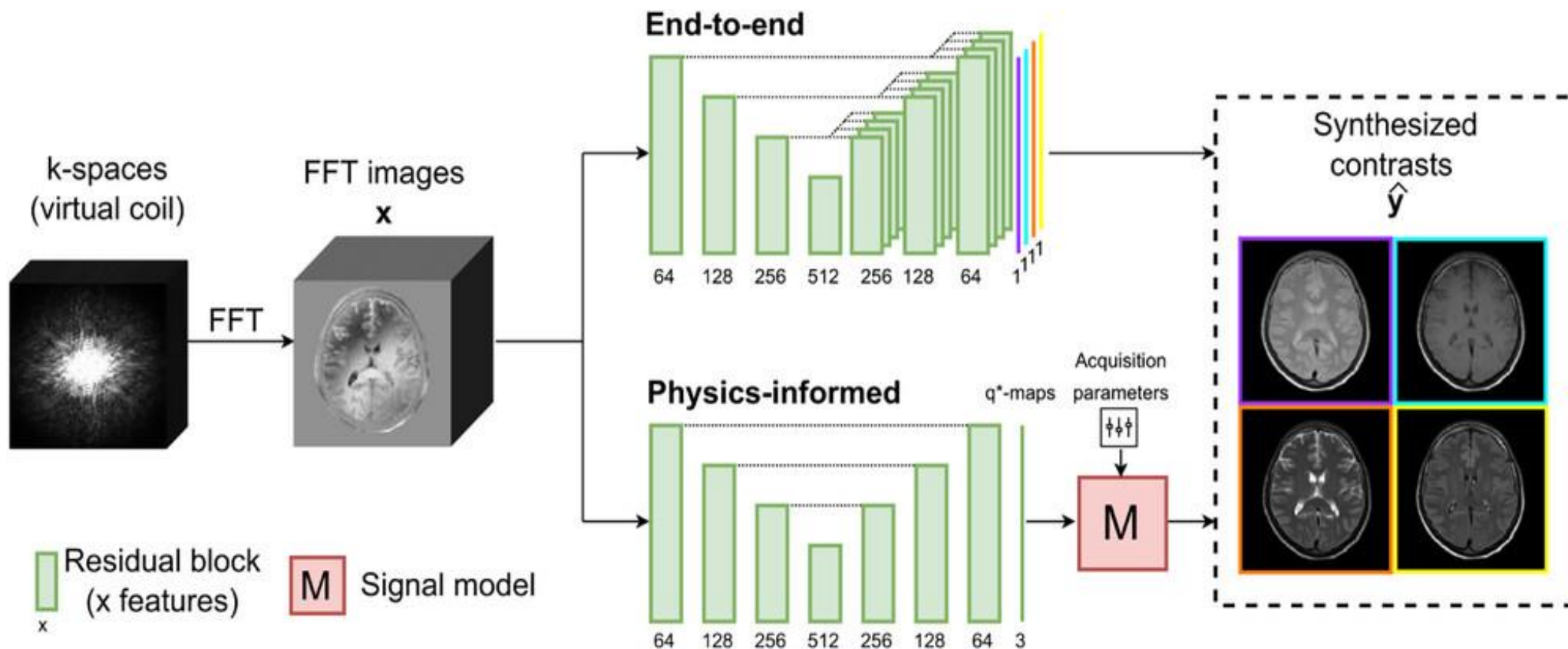
MIART TODAY

Li et al. — DoMa-Seg: Dose Map Guidance for Medical Image Segmentation in Head and Neck Radiotherapy — Oral 1: a dose map **is** the physics, used as guidance rather than as a metric.

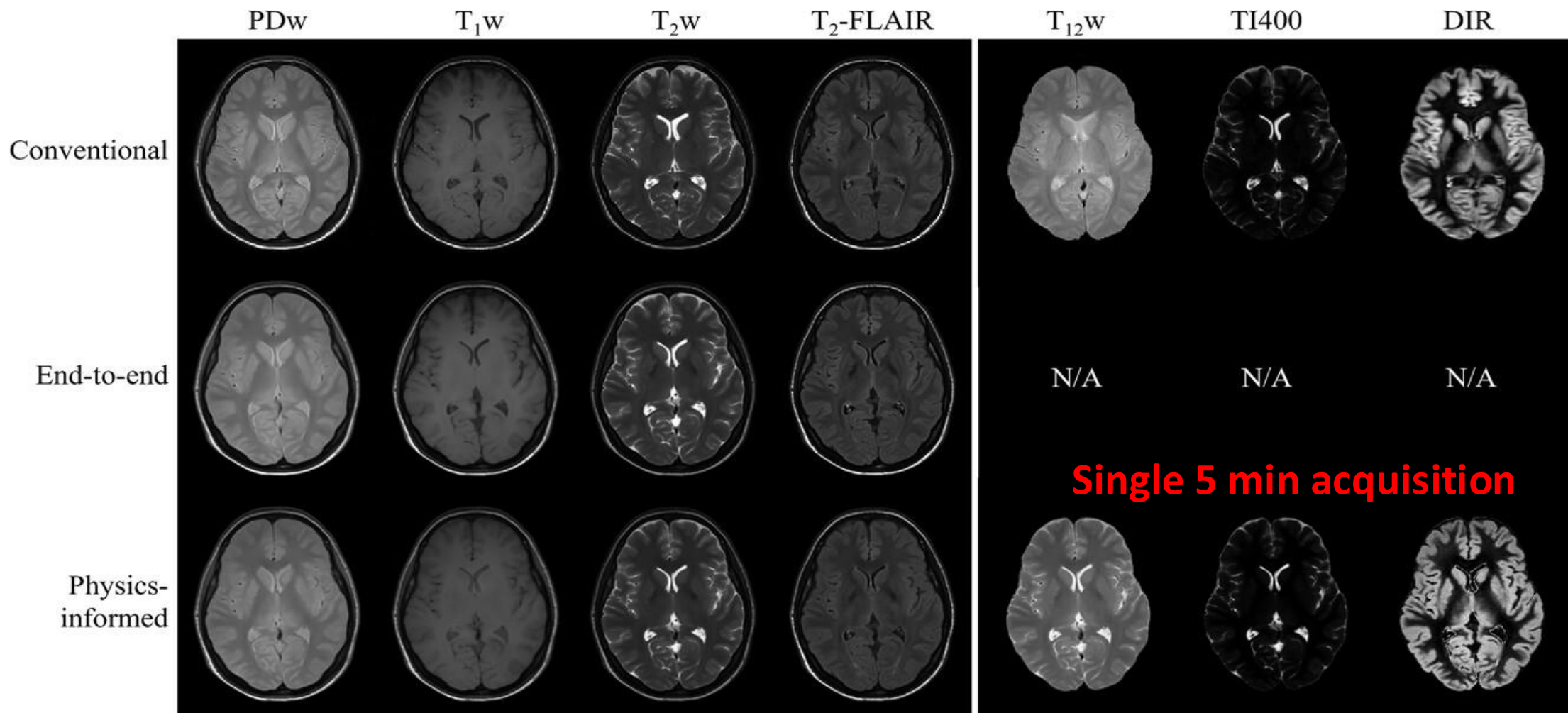
T-AM-001 · Li et al. — PhyRadGS: Physics-Aware Radiative Gaussian Splatting for Fly-Scanning CBCT Reconstruction · physics in the reconstruction, one step upstream

Quantitative MRI

with physics



Quantitative MRI with physics



Attempt 5: protocol agnosticism

train on contrasts that do not exist

Domain randomisation: synthesise wildly varied contrast during training so the unseen sequence is no longer unseen

Tested on a sequence held out of training entirely (FLAIR):

baseline, FLAIR unseen

106 HU

+ domain randomisation

99 HU

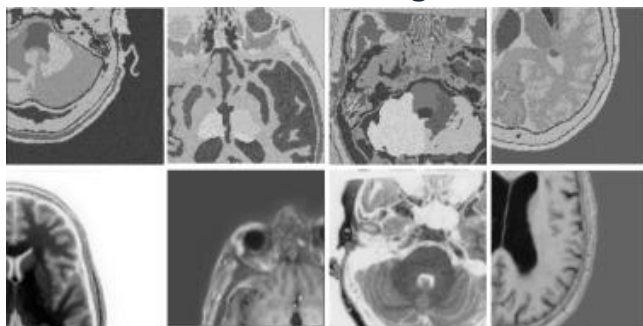
FLAIR in the training set

72.6 HU

mean absolute error on the held-out sequence

It helps. It does not close the gap. Be honest about that.

Worth it anyway: it reduce the need to retrain every time a protocol changes, which is the actual clinical cost.



Attempt 6: federate the world

Federated learning for sCT is now demonstrated, not hypothetical

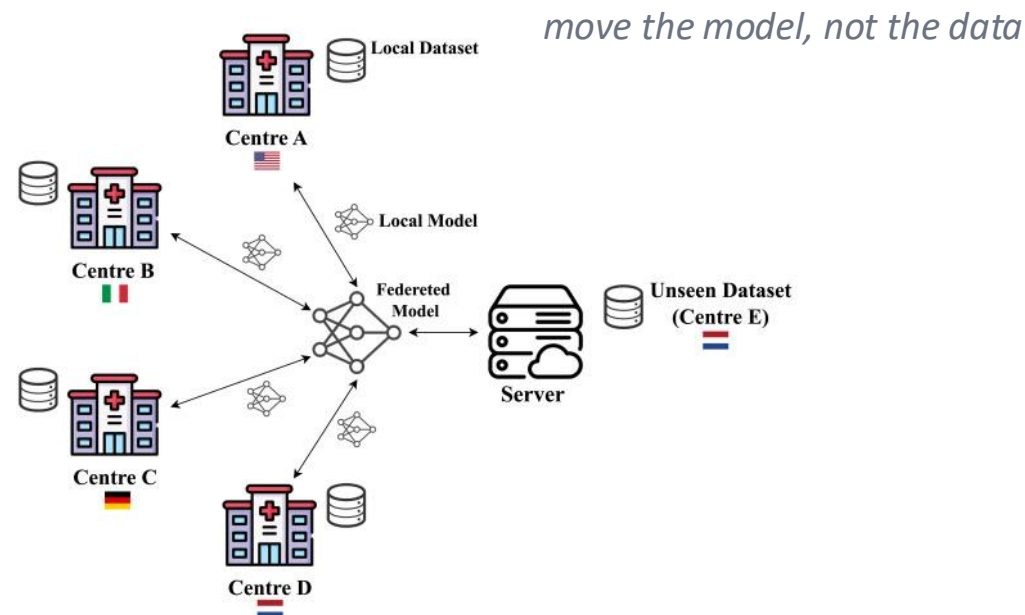
- four centres across Europe and the US trained a shared U-Net
- no images left any institution
- tested on a fifth centre that was never in the federation

Held-out centre: median MAE 102 HU · SSIM 0.89

Privacy is the problem federated learning solves. Protocol heterogeneity is the problem it does not.

It is a governance answer to a physics question — necessary, and on its own not sufficient.

And it does not reduce the validation burden at all: every participating centre still has to commission the result locally.



And foundation models?

the question you were expecting

Plausibly yes

large pre-trained encoders already beat hand-crafted features for RT outcome prediction

general segmentation models cover 100+ structures out of the box

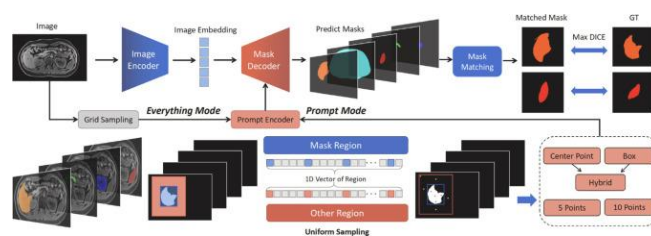
prompting is now text, not just points and boxes

But look what it costs

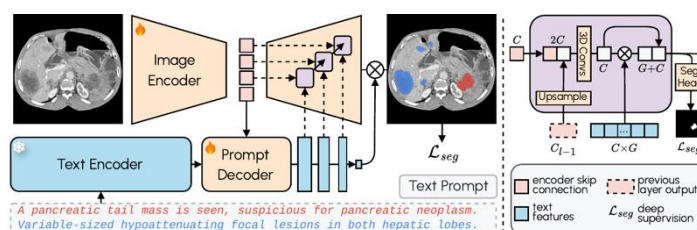
a general model has a general intended use — and intended use is what gets certified

more tasks, more anatomies, more failure modes to enumerate

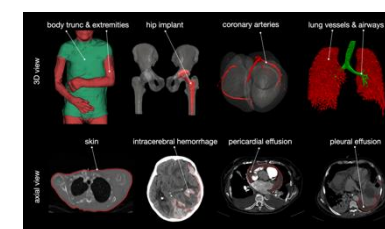
no RT-specific certification; the best-known general segmentation tool has none at all



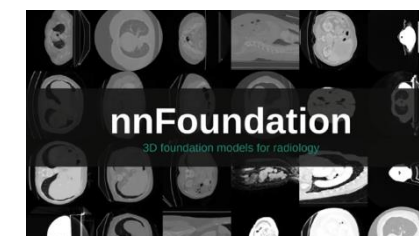
SAM, prompt-based



VoxTell, free-text prompts



TotalSegmentator, 100+ structures



nnFoundation

A model that does everything makes clinical evaluation combinatorially worse

So the interesting research problem is not the foundation model, but how to speed up evaluation that makes one certifiable

MIART TODAY

Megne Choudja et al. — Motion-Consistent Memory for Foundation Model-Based Tumor Tracking in MR-Guided Radiotherapy — Oral 3: a foundation model doing a real RT job, with memory.

M-PM-004 · Yamamoto & Kikuchi — TotalFM: An Organ-Separated 3D-CT Foundation Model Leveraging Large-Scale Routine Clinical Radiology Data

Take away 3

Where we could better focus

Uncertainty maps correlate with dose error and nobody ships them

A downstream task can QA a synthesis when no ground truth exists

Reconstruction and synthesis are already coupled — commission them as one

Physics-informed models attack domain shift at the root

Domain randomisation and federation help, and neither closes the gap

Every attempt on this list converges on the same missing piece: an automatic, per-case statement of whether to trust this image

4

QUESTION 4 — FUTURE DIRECTIONS

What comes next?

Some bets, in decreasing order of confidence

≈ 4 minutes

Synthesis under a time budget

Offline, the constraint is accuracy. Online, it is accuracy within the time the patient is on the couch

What that changes

- inference time becomes a hard clinical constraint, not a table entry
- diffusion's sampling cost stops being an implementation detail
- QA has to happen inside the same window — nobody reviews offline
- the model cannot defer to a human who is not there

The in-loop confidence check is the enabling technology, not a bigger model

Synthesis as a part of an agent

Synthesis is here to stay as a cooperating module in the workflow of the future

- synthesis as an independent technique to perform QA
- instead of choosing approach computation will become so cheap that we will run all of them and benchmark the results

Synthesis in a multi-task model to obtain a multi-pass on the fly evaluation

b. One-for-All Adaptive Radiotherapy Planning Agent

A query describes the needs

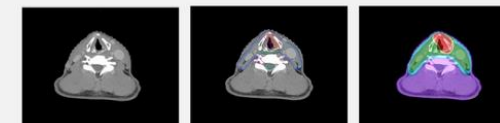


Anatomy changed. Check whether original plan still meets target coverage and OAR constraints; adaptive replan likely needed.

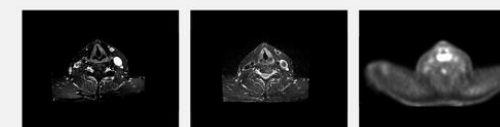
Daily CBCT



+



Planning Images
(CT/4DCT + Contour + Dose)



Diagnostic Images
(MRI/PET-CT)

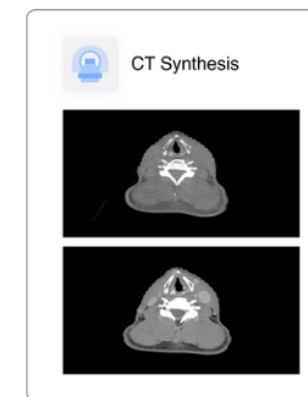
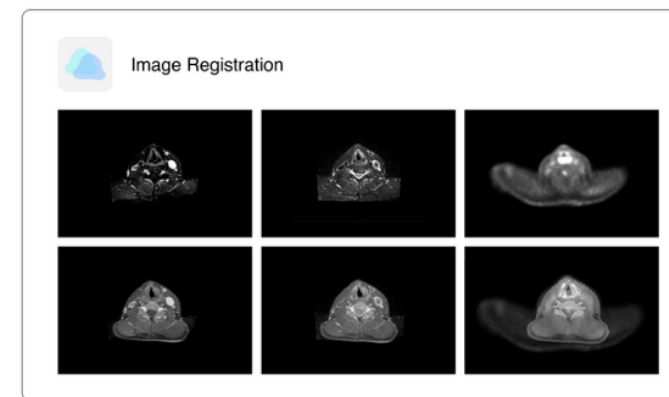
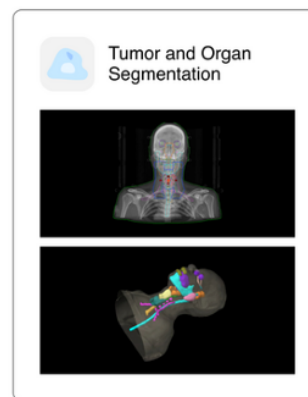
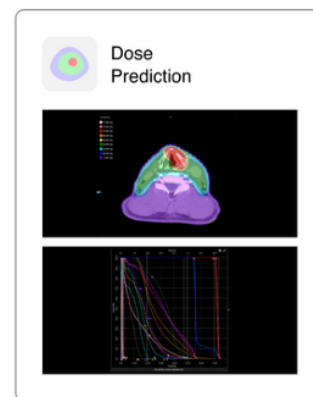
01

03

02

05

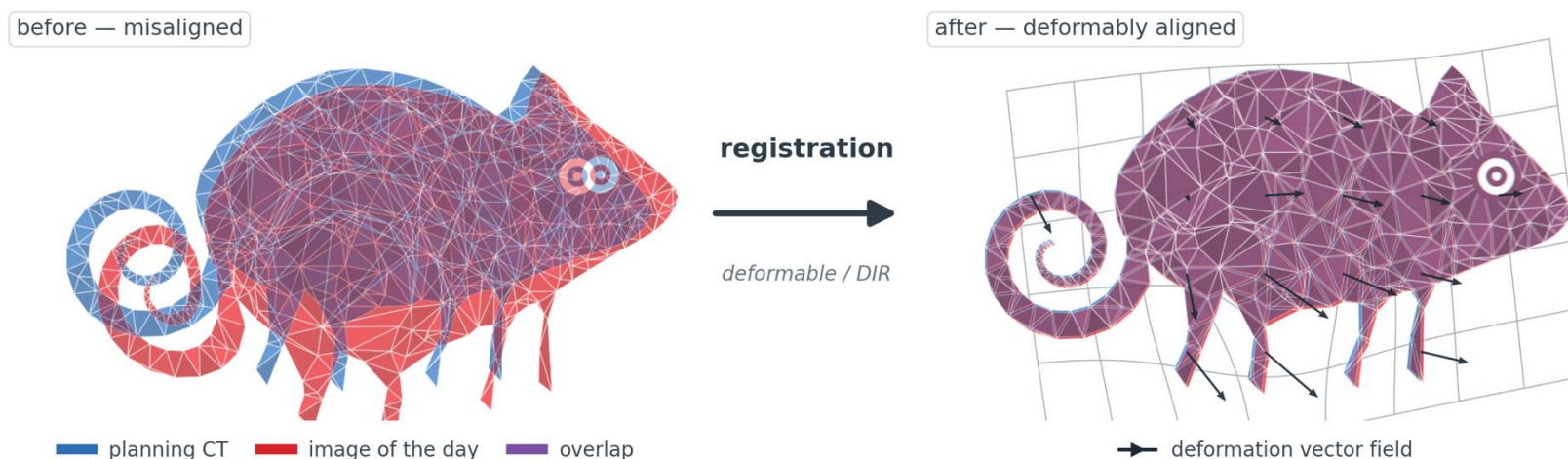
04



Synthesis as a service of another task

Register

two chameleons, one anatomy — deformably matched



Cross-modality registration is hard because the intensities are not comparable

Synthesise one modality into the other's space, then register monomodal:

→ the problem gets easier, not the model bigger

And here the synthetic image never touches the dose calculation, so the trust bar is lower and adoption is faster

Do we need the synthetic image at all?

The sCT is an intermediate representation, not the goal

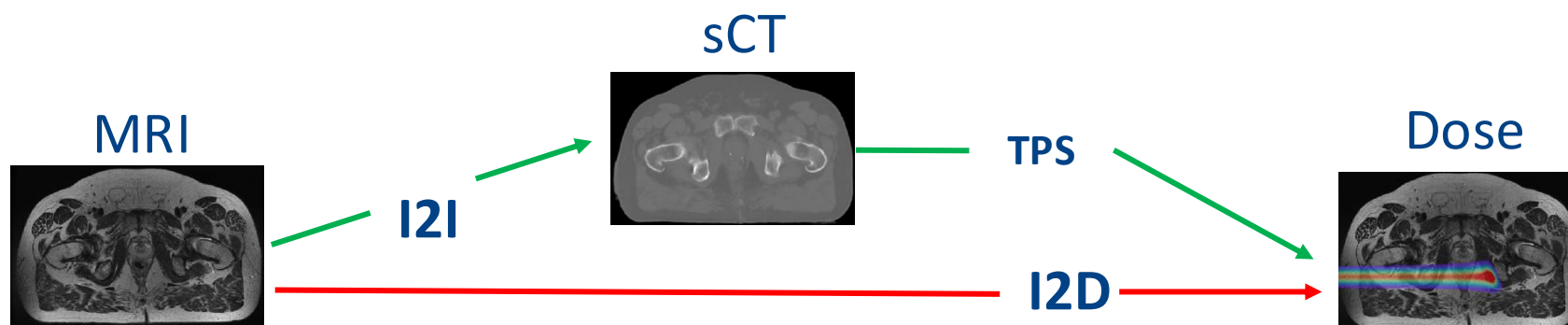
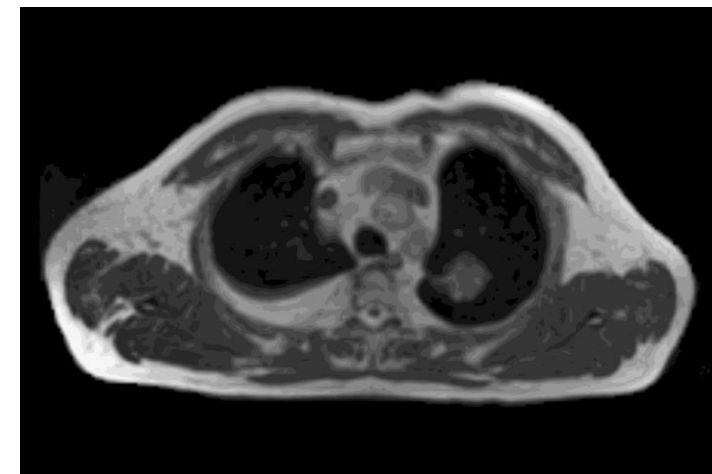
- the goal is a dose distribution
- direct MRI-to-dose has been shown for photons and for protons

What you gain

- no HU intermediate to be wrong about
- fewer steps to commission

What you lose is the thing a physicist can look at

Deleting the human-inspectable intermediate also deletes the QA surface



I2I: image to image I2D: image to dose

The speculative one

Everything so far synthesises anatomy. The interesting target is biology

- PET tracer uptake
- hypoxia and perfusion maps
- functional MRI-derived response signals

The pull is dose painting: prescribing to biology rather than to a contour

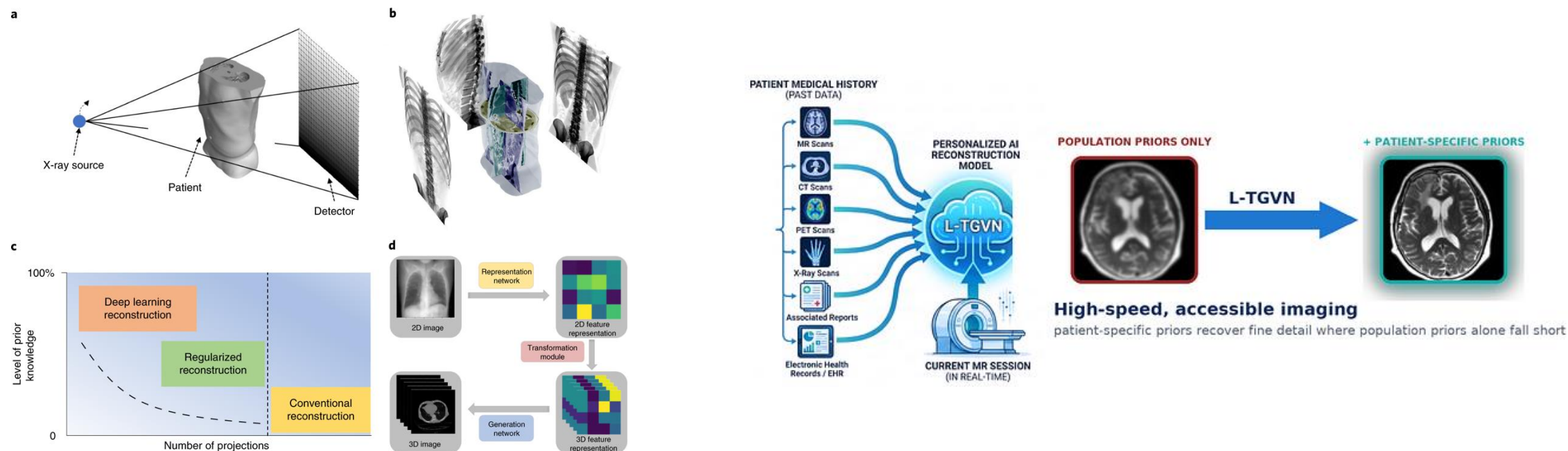
Be honest about this one: there is no validation path yet

You cannot verify a synthetic biological map against a dose calculation

→ The whole trust argument in this talk does not transfer

And on the reconstruction side

priors get personal



more prior knowledge, fewer projections — the trade the whole field is making

and in RT the prior can be this patient, from this week

Population priors are what learned reconstruction uses today

In RT we have something diagnostic imaging usually does not: prior scans of this patient, from this week

MIART TODAY

Jaouen et al. — Cross-Modal Spatial Gating of Planning-MRI Priors for Longitudinal CBCT-to-CT Synthesis — Oral 2: the planning MRI as the prior for today's CBCT. Exactly this argument.

How the field will actually move

one sentence

**Shared data and shared metrics & tools to calculate them
moved this field further than any architecture did**

SynthRAD2023 / 2025

3400+ datasets, 5 centres,
fully open pipeline

MESCAL

benchmark dataset and
acceptance levels

DoseRAD2026

this MICCAI — dose
prediction, same model

The answers are in this room

MIART, today

Nine oral papers. I have just spent thirty minutes on the problems; these are the people with the attempts.

Oral 1 · 08:45 · segmentation, contour quality & calibration

Fernandez Salamanca et al.	Recognizing Dosimetrically Irrelevant Edits of Auto-Segmentations in Head and Neck Organs of Interest	constraint 1 — which errors matter
Li et al.	DoMa-Seg: Dose Map Guidance for Medical Image Segmentation in Head and Neck Radiotherapy	attempt 4 — dose as guidance
Rangnekar et al.	Parameter-Efficient Pretrained-CT-to-MRI Transfer for Rectal Cancer Segmentation: Performance-Calibration Trade-Offs	constraint 4 — transfer vs calibration

Oral 2 · 10:30 · cross-modality image synthesis

Guija-Valiente et al.	SynthRCT: Scalable Conditional Deformation Synthesis for Synthetic Repeat CT Generation	constraint 3 — make the paired data
Rodriguez-Gonzalez et al.	Can We Trust Synthetic CT Algorithms? Uncertainty-Aware Evaluation of CBCT to CT Synthesis for Adaptive Proton Therapy	attempt 1 — uncertainty as QA
Jaouen et al.	Cross-Modal Spatial Gating of Planning-MRI Priors for Longitudinal CBCT-to-CT Synthesis	bet 4 — priors get personal

Oral 3 · 11:45 · adaptive radiotherapy

Cubero et al.	CBCT Segmentation in Head and Neck ART: A Longitudinal Deep Learning Approach	attempt 2 — downstream-task QA
Megne Choudja et al.	Motion-Consistent Memory for Foundation Model-Based Tumor Tracking in MR-Guided Radiotherapy	foundation models, doing a real job
El Aichi et al.	Uncertainty-Aware Out-of-Field Dose Prediction in External Beam Radiotherapy	attempt 1 — uncertainty on dose



Every one of my complaints has somebody in this room working on it: super cool!

And across the corridor

MICCAI 2026, this week

The four placeholders I could not fill last year are filled by this year's main conference.

- **Hallucination on the hard cases**
T-AM-093 · Zala et al. — Attention is Matter for Inclusiveness: Generating Synthetic CT for Patients with Hip Implants
- **MR-only synthesis, still moving**
T-AM-019 · Li et al. — DPMix-GAN: Dual-domain Perception and Dynamic Patch-wise Mixing GAN for 3D CT Synthesis in MR-only Radiotherapy
- **Domain shift, at test time**
T-AM-109 · Iele et al. — Multi-Agent Test-Time Adaptation for Robust Medical Image-to-Image Translation
- **Domain shift, by restyling**
T-PM-123 · Liu et al. — IntraStyler: Intra-Domain Style Synthesis for Cross-Modality MRI Domain Adaptation
- **Calibrated uncertainty, with guarantees**
M-PM-105 · Tan et al. — Conformal 3D Lesion Segmentation with Balanced Risk Control
- **Knowing you are out of distribution**
W-PM-180 · Das et al. — PROTON: Prototype-Based Test-Time Online OOD Detection for Medical VLMs
- **Reconstruction and the task, jointly**
T-AM-057 · Ki et al. — DosePlug: Observation-Driven Dose Modeling for Robust Low-Dose Reconstruction and Segmentation
- **Physics inside the reconstruction**
T-AM-001 · Li et al. — PhyRadGS: Physics-Aware Radiative Gaussian Splatting for Fly-Scanning CBCT Reconstruction
- **A CT foundation model**
M-PM-004 · Yamamoto & Kikuchi — TotalFM: An Organ-Separated 3D-CT Foundation Model Leveraging Large-Scale Routine Clinical Radiology Data

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Three things

if you remember nothing else

1

Generalisation, not fidelity

The architecture race is over. Domain shift across scanner, protocol and site is the open technical problem.

2

The last mile is trust, let's try to facilitate it

Image-similarity metrics do not predict dosimetric ranking. What gets deployed is what a physicist can test and sign for.

3

Automate the per-case check

Uncertainty, downstream-task QA, coupled commissioning — every attempt converges here, and foundation models make it urgent.

AN INVITATION

Let's keep arguing — for two days

35 minutes only gets you the questions. A workshop gets you the disagreement.



AIinRT 2027

Artificial Intelligence in Radiotherapy



Everything in the questions is a workshop session waiting to happen

Come and tell me which of my questions were the wrong ones

;))



1–2 April 2027

two full days

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up to 200 people

workshop format, not a lecture series

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Abstract submission closes **1 December 2026**

Six 90-min sessions · Invited state-of-the-art talks (30 min) · Proffered papers (9+3 min)
A cross-disciplinary keynote each day · Double-blind review

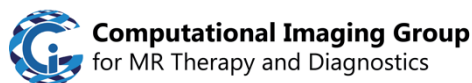
Sessions

- S1** Segmentation & Registration
- S2** Reconstruction & Synthesis
- S3** Foundation Models, Text, Explainability & Uncertainty
- S4** Dose & Adaptive Workflows
- S5** Clinical Predictions & Outcomes
- S6** Implementation, QA & Ethics



<https://AlinRT.org> — Call for Abstracts opens on 1 September

Organized by



In collaboration with



Thank you

Questions welcome — and I would rather have the hard ones

m.maspero@umcutrecht.nl



Slides available until 30 October at
<https://surfdrive.surf.nl/s/Aago8teqPRp7jep>



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Artificial Intelligence in Radiotherapy



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